

PAPER

Neural ARFIMA model for forecasting BRIC exchange rates with long memory under oil shocks and policy uncertainties

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Abstract

Accurate forecasting of exchange rates remains a persistent challenge, particularly for emerging economies such as Brazil, Russia, India, and China (BRIC). These series exhibit long memory, nonlinearity, and non-stationarity properties that conventional time series models struggle to capture. Additionally, there exist several key drivers of exchange rate dynamics, including global economic policy uncertainty, US equity market volatility, US monetary policy uncertainty, oil price growth rates, and country-specific short-term interest rate differentials. These empirical complexities underscore the need for a flexible modeling framework that can jointly accommodate long memory, nonlinearity, and the influence of external drivers. To address these challenges, we propose a Neural AutoRegressive Fractionally Integrated Moving Average (NARFIMA) model that combines the long-memory representation of ARFIMA with the nonlinear learning capacity of neural networks, while flexibly incorporating exogenous causal variables. We establish theoretical properties of the model, including asymptotic stationarity of the NARFIMA process using Markov chains and nonlinear time series techniques. We quantify forecast uncertainty using conformal prediction intervals within the NARFIMA framework. Empirical results across six forecast horizons show that NARFIMA consistently outperforms various state-of-the-art statistical and machine learning models in forecasting BRIC exchange rates. These findings provide new insights for policymakers and market participants navigating volatile financial conditions. The `narfima R` package provides an implementation of our approach.

Key words: Forecasting; Long memory processes; Neural networks; Asymptotic stationarity; Conformal prediction

1. Introduction

Exchange rates significantly influence macroeconomic outcomes, shaping trade balances, capital flows, inflationary pressures, and financial stability, thereby becoming a central focus for policymakers, central banks, and market participants in an increasingly interconnected global economy (Eichengreen, 1998; Hausmann et al., 1999; Stoica and Ihnatov, 2016). Their role in assessing countries' financial stability has long been established (Taylor, 2001), with accurate forecasts proving indispensable for guiding monetary policy, designing capital controls, and implementing macro-prudential measures (Wieland and Wolters, 2013; Lubik and Schorfheide, 2007). For commodity-dependent economies, where exchange rate fluctuations directly impact inflation forecasts and broader economic stability, reliable projections are particularly vital (Rossi, 2013). Within this context, spot exchange rates, reflecting current market prices for immediate currency delivery, are especially important as they directly influence trade prices, international capital flows, external debt obligations, and monetary policy decisions (Pilbeam and Langeland, 2015). Unlike aggregate measures such as the real effective exchange rate, spot rates respond swiftly to short-term market conditions, policy shifts, and uncertainty shocks, with their continuous trading facilitating rapid assimilation of new information (Bartsch, 2019). This makes spot markets an ideal setting for analyzing the transmission of oil price shocks and policy uncertainty to exchange rate dynamics across multiple horizons. Given the rising global prominence of the BRIC economies¹, collectively accounting for 37.3% of world GDP (PPP) and a rapidly growing share of global trade (O'Neill, 2011; Hopewell, 2017; Sengupta et al., 2025; Nasir et al., 2018), understanding and accurately forecasting spot exchange rates in these economies has become increasingly critical for macroeconomic planning, financial stability, and policy coordination.

¹ As of January 1, 2024, the BRICS has expanded to include eleven members (Brazil, Russia, India, China, South Africa, Argentina, Egypt, Ethiopia, Iran, Saudi Arabia, and the United Arab Emirates). This expansion has increased the collective share of global GDP. Further details can be found at: <https://www.weforum.org/stories/2024/11/brics-summit-geopolitics-bloc-international/>.

Although emerging economies such as the BRIC nations have become increasingly influential in the global economy, they remain particularly vulnerable to global shocks, often experiencing rapid and severe currency fluctuations than developed nations. This increased susceptibility stems from greater exposure to external shocks, fragile financial markets, and risk of sudden reversals in capital flows, a phenomenon termed “flight-to-quality” (Bernanke et al., 1994; Calvo and Talvi, 2005). In response to these challenges, extensive literature has emphasized the importance of incorporating various forms of uncertainty measures, such as economic policy uncertainty (EPU) and geopolitical risks (GPR), into exchange rate forecasting frameworks (Kumar et al., 2024; Salisu et al., 2022). From a theoretical perspective, when domestic uncertainty exceeds foreign uncertainty, domestic investors tend to invest in foreign currency assets, triggering exchange rate movements (Balcilar et al., 2016). Moreover, economic uncertainties affect expectations about costs and returns, which in turn influence both supply and demand in currency markets (Benigno et al., 2012). Motivated by these theoretical foundations, recent empirical studies have focused on modeling the interplay between EPU and exchange rate dynamics. For instance, Zhou et al., 2020 demonstrated the enhanced forecasting performance of Generalised Autoregressive Conditional Heteroskedasticity - Mixed Data Sampling (GARCH-MIDAS) models incorporating Sino-US EPU in predicting Chinese exchange rate volatility, while Benigno et al., 2012 explored how monetary, inflation, and productivity uncertainties influence real exchange rates. Further studies have consistently confirmed the robust predictive power of EPU in emerging markets across short and long horizons (Colombo, 2013; Sin, 2015; Juhro and Phan, 2018; Abid, 2020). Alongside EPU, other financial uncertainty indices such as the US Equity Market Volatility (EMV) and US Monetary Policy Uncertainty (MPU) measure distinct aspects of the economic environment that potentially impact currency markets. Empirical evidence from Mueller et al., 2017 indicates that US MPU significantly affects currency risk premia, with emerging market currencies exhibiting greater sensitivity due to “reach for yield” behavior. Additionally, Istrefi and Mouabbi, 2018 documents an inverse relationship between US MPU and economic activity, where increased US MPU typically prompts dollar appreciation driven by safe-haven demand during financial stress. Collectively, these findings underscore the importance of incorporating multifaceted uncertainty measures to better capture the complex dynamics governing exchange rates in emerging economies.

In addition to uncertainty measures, other external economic factors, such as the relationship between oil prices and exchange rates, form a critical nexus in international finance, particularly for economies with significant oil exposure. For instance, Beckmann et al., 2020 demonstrated that the oil price and exchange rate relationship exhibits time-varying characteristics depending on the nature of the underlying shock driving oil prices. The distinction between oil-exporting and oil-importing countries is fundamental, as highlighted by Chen et al., 2024, who found that exchange rate-oil price connectedness intensifies during crisis periods, with stronger transmission channels evident in oil-exporting economies. This finding is especially relevant for BRIC countries, which span the spectrum from major oil exporters (Russia, Brazil) to significant importers (China, India). Alongside oil price dynamics, short-term interest rates play crucial roles in exchange rate movements. In a recent study, Andrieş et al., 2017 examined this relationship in Romania using a structural vector autoregressive approach and uncovered bidirectional causality between interest rates and exchange rates, with the strength of the relationship varying across different time horizons. In a similar direction, Saraç and Karagöz, 2016 investigated the Turkish economy and found that short-term interest rates significantly impact exchange rate volatility, with the relationship intensifying during periods of economic distress. Their analysis further highlighted that interest rate differentials serve as a more reliable predictor of exchange rate movements than absolute interest rate levels. For BRIC economies, which exhibit varying degrees of capital account openness and monetary policy independence, the interest rate channel represents a critical transmission mechanism through which both domestic and external shocks propagate to exchange rates. Thus, when combined with oil price dynamics and uncertainty measures, they can capture the complex dynamics governing exchange rates in emerging economies.

The complexity of exchange rate dynamics in emerging markets makes forecasting challenging, motivating the development of models beyond standard benchmarks. However, Meese and Rogoff, 1983 demonstrated that simple random walk models often outperform more advanced econometric approaches in out-of-sample forecasts. Numerous efforts have been made to improve the accuracy of exchange rate predictions, the existing literature encompasses a wide range of strategies, including Taylor rule-based fundamentals (Molodtsova and Papell, 2009), nonlinear methods (Kilian and Taylor, 2003), and Kalman filter-based models (Date and Maunthrooa, 2025). More recently, with advances in data-driven techniques and the increasing availability of macroeconomic datasets, the adoption of statistical and machine learning techniques for exchange rate forecasting has surged (Plakandaras et al., 2015). For instance, Ngan, 2016 employed an autoregressive integrated moving average (ARIMA) model to capture the exchange rate dynamics in Vietnam, while Karemera and Kim, 2006 revealed that an autoregressive fractionally integrated moving average (ARFIMA) framework can outperform the random walk method in forecasting exchange rates for several developed economies. Similarly, Pilbeam and Langeland, 2015 investigated the effectiveness of the univariate GARCH model in forecasting foreign exchange market volatility, and Galeshchuk, 2016 explored the use of neural networks in forecasting exchange rates across multiple currencies. More recently, deep learning approaches have been applied to forecast exchange rates in emerging economies (Abir et al., 2024); however, these algorithms require high-frequency data, limiting their effectiveness under data scarcity, and they struggle to capture the long-memory characteristics of exchange rates. Despite the methodological advances, a majority of these studies have predominantly focused on exchange rates of developed countries and OECD nations, largely overlooking the distinct dynamics of emerging economies and their interactions with macroeconomic drivers. To address this gap, we systematically examine how multiple uncertainty measures, such as global EPU (GEPU), US EMV, and US MPU, alongside oil price growth rates and country-specific short-term interest rate differentials, impact BRIC exchange rate dynamics through causal analysis techniques. Given the nonlinear characteristics of the data, we employ a nonlinear Granger causality test to identify the key drivers. Furthermore, we detect that exchange rate series of the BRIC economies exhibit structural complexities, including long-term memory, nonlinearity, and non-stationary behavior that traditional time series forecasting approaches, such as linear ARIMA or ARFIMA models, fail to capture. Although advanced neural networks can model the inherent nonlinearities in the exchange rate series, they often lack the theoretical foundations required to model long-term memory and non-stationary patterns.

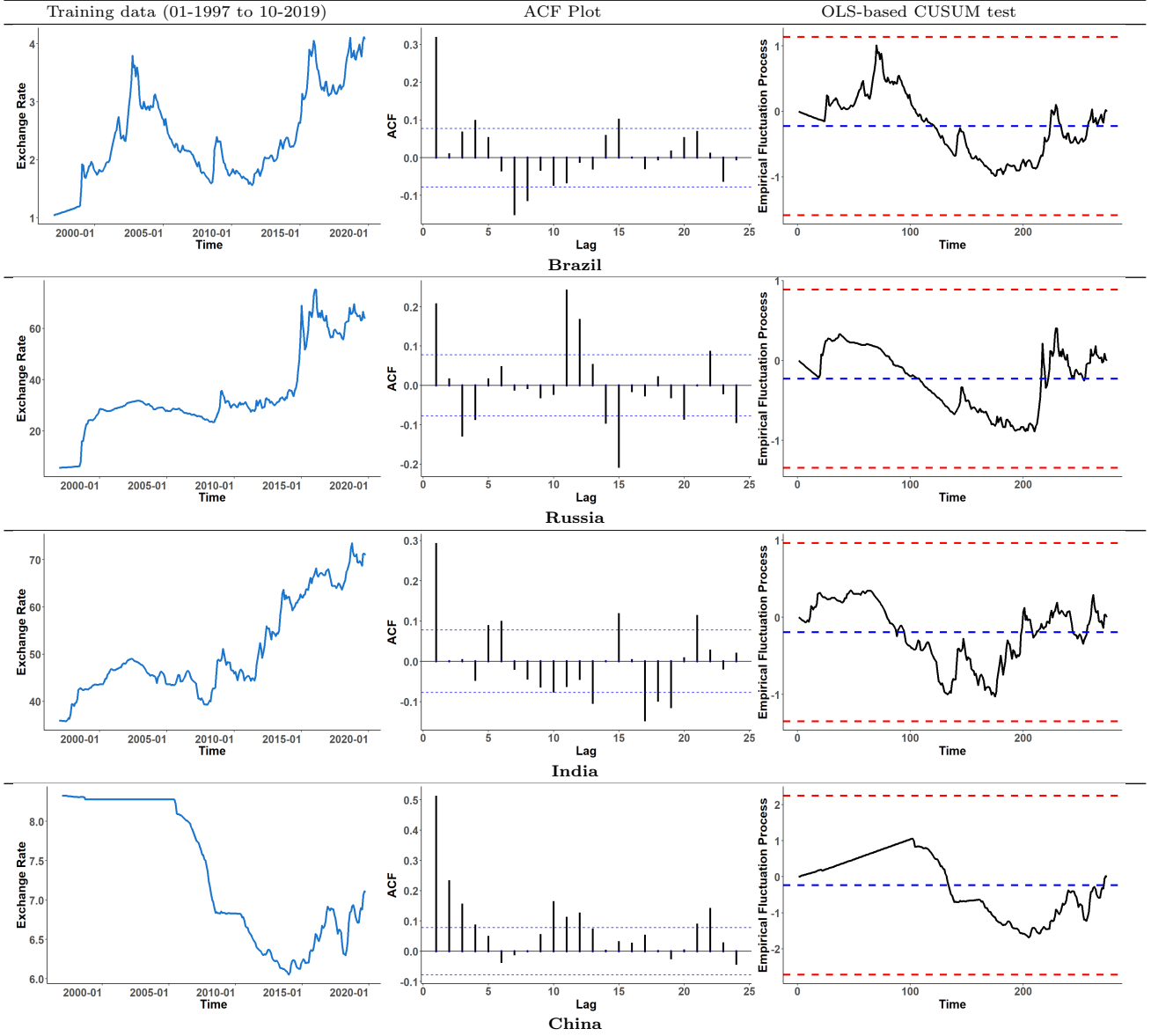
To address these methodological limitations, we propose the Neural AutoRegressive Fractionally Integrated Moving Average (NARFIMA) model, an ensemble approach that integrates the strengths of classical statistical methods with advanced machine learning techniques. The mechanism of the proposed NARFIMA model operates through a structured two-stage procedure. In the initial phase, a linear ARFIMA model is fitted with lagged observations of the exchange rate series, auxiliary uncertainty measures, and macroeconomic drivers to produce in-sample residuals. These ARFIMA model residuals capture the unexplained variations in the exchange rate dataset after accounting for linear long-memory effects and key economic drivers. In the subsequent step, the residuals are combined with the exchange rate time series data, uncertainty measures, and macroeconomic drivers, and are then modeled using an autoregressive neural network architecture to capture nonlinear dependencies and underlying structural patterns. This ensemble framework can capture long-term memory, nonlinear dependencies, and complex interactions between exchange rates and multiple economic drivers. Furthermore, the asymptotic stationarity and geometric ergodicity conditions obtained through Markov chain analysis confirm the theoretical robustness of the NARFIMA model; we have empirically verified that the assumptions necessary for these results are satisfied. These theoretical properties have several economic implications and policy relevance, as central banks and financial institutions rely on forecasting models that are asymptotically stable, interpretable, and reliable. Empirical evaluations conducted on the BRIC economies' exchange rates demonstrate that the NARFIMA model substantially enhances predictive accuracy and robustness, surpassing state-of-the-art statistical and machine learning forecasting frameworks. Its superior performance in capturing complex nonlinear interactions between exchange rate and causal drivers while modeling the long-term memory and non-stationary patterns of the endogenous variable underscores the practical relevance and reliability of our proposed approach. Overall, the integration of theoretical rigor with empirical validation positions NARFIMA as an effective forecasting technique, particularly suited for analyzing the complex exchange rate dynamics under varying policy uncertainties and critical macroeconomic shocks. By capturing both long-term memory and complex nonlinear dynamics between uncertainty measures, short-term interest rate differentials, oil shocks, and exchange rates, our approach provides a comprehensive tool that can deliver valuable insights for policymakers and researchers operating in an increasingly uncertain global economic environment. To further enhance its practical relevance, we perform uncertainty quantification using conformal prediction, which can be integrated with the NARFIMA framework and is vital for policy applications.

The remainder of this paper is structured as follows. Section 2 offers a detailed description of the data characteristics. Section 3 introduces the proposed NARFIMA methodology, detailing its architecture and theoretical properties, including asymptotic stationarity of the NARFIMA model. In Section 4, we present the causality analysis, empirical validation of theoretical conditions, and performance evaluation comparing the forecasting accuracy of the NARFIMA model with sixteen benchmark methods across six forecast horizons. Robustness checks and statistical significance tests supplement the evaluation. Uncertainty quantification through conformal prediction and ablation study is presented in Section 5. Policy implications arising from our findings, relevant for central banks and international investors, are discussed in Section 6. Finally, Section 7 concludes this study and outlines future research directions.

2. Data Characteristics

In this study, we analyze monthly spot exchange rates and several macroeconomic covariates for the BRIC economies from January 1997 (1997-01) to October 2023 (2023-10). Our analysis employs a rolling window approach with six forecast horizons: short-term (1- and 3-month-ahead), semi-long-term (6- and 12-month-ahead), and long-term (24- and 48-month-ahead) for each country. For short-term forecasting, with 1-month-ahead and 3-month-ahead horizons, the training periods span from 1997-01 to 2023-09 and 1997-01 to 2023-07, while the test horizons are 2023-10 and 2023-08 to 2023-10, respectively. In the case of semi-long-term forecasting, which includes 6-month-ahead and 12-month-ahead horizons, the training periods extend from 1997-01 to 2023-04 and 1997-01 to 2022-10, with corresponding test horizons from 2023-05 to 2023-10 and 2022-11 to 2023-10, respectively. Finally, for long-term forecasting with 24-month-ahead and 48-month-ahead horizons, the training period ranges from 1997-01 to 2021-10 and 1997-01 to 2019-10, while the test horizons cover 2021-11 to 2023-10 and 2019-11 to 2023-10, respectively. The target variable, spot exchange rate of BRIC countries, is obtained from the Federal Reserve Economic Data (FRED) repository². Monthly exchange rate values are calculated as the averages of daily exchange rates based on noon buying rates in New York City for foreign currency cable transfers and are seasonally unadjusted. Table 1 presents the time series plots of exchange rate dynamics, along with their autocorrelation functions (ACF) plots for the BRIC nations. We observe that all BRIC currencies, except the Chinese yuan, have depreciated against the USD over time. China maintained a currency peg until 2005, and even after shifting away from the peg, its central bank continued active intervention in the exchange rate market. In contrast, the other BRIC nations follow a floating exchange rate system. Additionally, a sharp spike in exchange rates is evident during the global recession, affecting all BRIC countries except China. On the other hand, the ACF plots for the BRIC nations indicate that the autocorrelation decays slowly and stays above the significance bounds for some lags, providing weaker evidence of long memory. Furthermore, to investigate possible structural breakpoints in the exchange rate series, we employ the ordinary least squares (OLS)-based CUSUM test, which helps account for regime shifts (Ploberger and Krämer, 1992). The test examines the cumulative sum of recursive residuals to identify any deviations from the model's stability, indicating potential structural breakpoints. Our implementation of the OLS-based CUSUM test shows that no statistically significant breakpoints were found in the exchange rate series for BRIC countries, as depicted in Table 1. The absence of significant breakpoints supports the conclusion that the exchange rate series has been relatively stable during the observed period, suggesting that regime shifts are unlikely to affect the forecasting models.

² <https://fred.stlouisfed.org/>.

Table 1. Training data for exchange rates of BRIC nations, alongside the ACF plot and OLS-based CUSUM test results.

Among the economic drivers, we consider four widely used news-based uncertainty measures³, namely GEPV, US EMV, US MPU, and GPR. An overview of their definitions and construction is provided in the Appendix B.1. Beyond uncertainty indices, we incorporate macroeconomic indicators such as oil prices, interest rates, and inflation. We consider the global price of West Texas Intermediate (WTI) crude oil (USD per barrel) and stabilize its variability by computing the oil price growth rates. The resultant series helps to capture the effect of oil price fluctuations on exchange rate movements over time. Additionally, we include short-term interest rates, which reflect the cost of borrowing for a duration under 24 hours between financial institutions or for government securities. These rates, measured in percentages for the BRIC economies and the US, serve as a key indicator of short-term financial system liquidity, responding to central bank interventions and market fluctuations. To assess relative monetary policy stance and capital flow dynamics, we compute the short-term interest rate differential (IRD) between the US and each BRIC economy. This country-specific short-term IRD significantly influences exchange rates, as higher BRIC interest rates tend to attract capital inflows, strengthening the domestic currency, while lower rates can lead to depreciation. We also consider CPI (Consumer Price Index) inflation rates for both the US and the BRIC economies, which measure price changes for a fixed basket of goods and services. The CPI inflation differential, reflecting the gap between US and country-specific CPI inflation rates, influences exchange rate movements, as higher domestic inflation indicates currency depreciation and vice versa. All macroeconomic indicators used in this analysis are collected from the FRED repository.

³ The historical dataset for all uncertainty indicators is obtained from <https://www.policyuncertainty.com/>.

We compute the summary statistics and analyze several global characteristics of the exchange rate datasets and auxiliary variables. Notably, we focus on seven key time series features: skewness, kurtosis, nonlinearity, long-range dependence, seasonality, stationarity, and outlier detection (Hyndman and Athanasopoulos, 2018). We employ Tsay’s and Keenan’s one-degree tests to assess nonlinearity, while long-range dependence is examined using the Hurst exponent. Unlike the ACF plots, which show finite-sample correlations, the Hurst exponent captures global scaling. Additionally, The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test is used to evaluate stationarity, while Ollech and Webel’s test is applied to detect seasonal patterns. The statistical characteristics of these datasets, summarized in Appendix B.2, reveal that most of the macroeconomic time series are non-stationary, except the oil price growth rate series, the CPI inflation series for Brazil and India, and the GPR series for Brazil and Russia. Most of the series do not exhibit seasonality, except for the GEPU and US EMV series. Additionally, nonlinear patterns are present in the majority of the dataset. The Hurst exponent values greater than 0.5 suggest persistent long-range dependence in exchange rate dynamics and all economic indicators. Moreover, to detect the presence of outliers in the dataset, we apply the Bonferroni outlier test using studentized residuals (Cook and Weisberg, 1982). Our analysis indicates that all series, except for the exchange rate series of Russia, exhibit significant outliers.

3. Neural ARFIMA Model

This section outlines the workflow of the proposed Neural ARFIMA (NARFIMA) model, designed for forecasting exchange rate dynamics in the BRIC economies. The NARFIMA framework integrates the strengths of the AutoRegressive Fractionally Integrated Moving Average with causal exogenous variables (ARFIMAx) to capture long-range dependencies in time series while leveraging a neural network to model complex nonlinear interactions within macroeconomic variables. The proposed model follows a sequential approach, where ARFIMAx is first employed to model exchange rate series based on historical observations and macroeconomic drivers. ARFIMAx captures short-term dependencies through its autoregressive and moving average components, while its fractional differencing mechanism accounts for long-term dependencies in exchange rate dynamics. The inclusion of macroeconomic indicators as exogenous variables further strengthens the model by incorporating external influences, providing a comprehensive representation of the exchange rate dynamics.

3.1. Model Formulation

Given T historical observations of exchange rate series $\{y_t; t = 1, 2, \dots, T\}$ and r macroeconomic drivers $\{X_{j,1}, X_{j,2}, \dots, X_{j,T}\}_{j=1}^r$, the ARFIMAx model is formulated as

$$\left(1 - \sum_{i=1}^{\tilde{p}} \tilde{\phi}_i B^i\right) (1 - B)^d y_t = \tilde{\mu} + \sum_{j=1}^r \tilde{\pi}_j B X_{j,t} + \left(1 + \sum_{k=1}^{\tilde{q}} \tilde{\theta}_k B^k\right) \epsilon_t, \quad (1)$$

where ϵ_t is white noise and B represents the backshift operator, such that $B y_t = y_{t-1}$ and $B X_{j,t} = X_{j,t-1}$. The parameters $\tilde{p}$ and $\tilde{q}$ denote the number of autoregressive and moving average terms, respectively, while $d \in (0, 0.5)$ represents the fractional differencing parameter responsible for capturing long-memory effects (Granger and Joyeux, 1980). The coefficients $\{\tilde{\phi}, \tilde{\theta}, \tilde{\pi}, \tilde{\mu}\}$ correspond to the autoregressive, moving average, exogenous components, and bias term, respectively. The fractional differencing operator, given by

$$(1 - B)^d = \sum_{v=0}^{\infty} \frac{\Gamma(v - d) B^v}{\Gamma(-d) \Gamma(v + 1)},$$

where $\Gamma(\cdot)$ denotes the gamma function, ensures that y_t is transformed into a stationary process. By allowing d to take non-integer values, this operator effectively captures long-term memory effects in time series, making it particularly well-suited for exchange rate datasets with slow decaying autocorrelation. Thus, the predictions $\{\hat{y}_t^{ARFIMA}\}$ generated from the ARFIMAx model by estimating the coefficients in Eqn. (1) capture the linear trajectory of the exchange rate series and the influence of auxiliary covariates. However, exchange rate series of emerging economies like BRIC often exhibit complex nonlinear structures and dynamic causal interactions, which the linear ARFIMAx model may fail to fully explain. Consequently, the residuals

$$e_t = y_t - \hat{y}_t^{ARFIMA}$$

of the ARFIMAx framework, captures the unexplained variations in exchange rate dynamics after accounting for linear long-memory effects. These residuals contain learnable structures with nonlinear dependencies and high-frequency fluctuations that cannot be effectively modeled using traditional parametric approaches. To address these limitations, the NARFIMA(p, q, k) framework integrates a feed-forward neural network to model complex nonlinear patterns.

The neural network component in the NARFIMA model is structured as a single hidden-layer architecture, enabling it to learn intricate nonlinear relationships between lagged values of exchange rate series, ARFIMAx residuals, and economic drivers. This single-layered framework offers a balance between computational efficiency and predictive power, restricting overfitting, and making it suitable for forecasting exchange rates in the presence of both linear and nonlinear dynamics. Due to limited data availability for macroeconomic modeling, highly computational deep learning models often fail to capture the data dynamics. On the contrary, NARFIMA utilizes an artificial neural network structure with only one hidden layer having k neurons; therefore, it does not overfit. The network is designed to receive inputs consisting of p historical exchange rate observations, q lagged values of ARFIMAx residuals, and one lagged value for each of the r macroeconomic covariates. The output of the network is a one-step-ahead

forecast of the exchange rate series, which is expressed as:

$$\hat{y}_{t+1} = f(y_t, y_{t-1}, \dots, y_{t-p+1}, e_t, e_{t-1}, \dots, e_{t-q+1}, X_{1,t}, X_{2,t}, \dots, X_{r,t});$$

where $t = \max(p, q), \dots, T$ and f represents the neural network function. By learning from historical data and residuals, the neural network effectively captures both linear and nonlinear relationships between input features and exchange rate dynamics, along with long memory dependence. In the proposed settings, two variants of the network are designed based on the inclusion or exclusion of skip connections. These skip connections allow input features to directly influence the output in addition to passing through the hidden layer, influencing the learning process and impacting model performance. The presence of a skip connection ensures that both linear and nonlinear components of the data are effectively integrated. The skip connection between the input and the output layer preserves the linear dynamics of the exchange rate series while allowing the hidden layer to learn the nonlinear interactions. Additionally, the presence of skip connections enhances training stability by mitigating the vanishing gradient problem and serves as a regularizing mechanism, which reduces overfitting by allowing direct information propagation and preventing unnecessary transformations. Thus, using the single-hidden layer neural network with a skip connection, the one-step-ahead forecasts of the exchange rate series can be generated as:

$$\begin{aligned} \hat{y}_{t+1} = & \mu_0 + \sum_{l=1}^k \mu_l \sigma \left(\tilde{\alpha}_l + \sum_{i=1}^p \beta_{i,l} y_{t-i+1} + \sum_{j=1}^q \gamma_{j,l} e_{t-j+1} + \sum_{m=1}^r \delta_{m,l} X_{m,t} \right) \\ & + \sum_{u=1}^p \phi_u y_{t-u+1} + \sum_{v=1}^q \eta_v e_{t-v+1} + \sum_{w=1}^r \zeta_w X_{w,t}, \end{aligned}$$

where k is the number of hidden nodes, $\{\tilde{\alpha}_l, \beta_{i,l}, \gamma_{j,l}, \delta_{m,l}\}$ are the connection weights between input and hidden layers, μ_l is the weight vector between hidden and output layers, $\{\phi_u, \eta_v, \zeta_w\}$ represent skip connection weights, μ_0 is the bias term, and $\sigma(\cdot)$ is the nonlinear activation function. The network weights are initialized randomly and trained using the gradient descent backpropagation approach (Rumelhart et al., 1986). In the variant of the neural network without skip connections, the mechanism follows a similar structure, but the skip connection weights $\{\phi_u, \eta_v, \zeta_w\}$ are set to zero, removing the skip connection between input and output layers. The above procedure generates a one-step-ahead forecast of the exchange rate series. For the multi-step ahead forecasts, we employ a recursive framework where the input layer is updated with the latest predictions at each step. Alongside point forecasts, the NARFIMA framework can be readily integrated with conformal prediction techniques to produce reliable prediction intervals (see Section 5).

3.2. Choice of Parameters

The NARFIMA model consists of primarily four tunable parameters, namely the number of lagged exchange rate observations (p), the number of historical values for ARFIMAx residuals (q), the number of nodes in the hidden layer (k), and the network structure indicating whether a skip connection is included (*skip*). To determine the optimal values of the parameters, we utilize a time series cross-validation strategy and select (p, q, k) by minimizing the root mean square error (RMSE) on the validation set ($\mathcal{V}$) as follows:

$$(p, q, k) = \underset{(p, q, k)}{\operatorname{argmin}} \sqrt{\frac{1}{|\mathcal{V}|} \sum_{t' \in \mathcal{V}} (y_{t'} - \hat{y}_{t'})^2},$$

where $y_{t'}$ and $\hat{y}_{t'}$ are the ground truth and forecasts generated by NARFIMA at time $t' \in \mathcal{V}$, respectively. For all the datasets, we build two different versions of the NARFIMA model, one with skip connections (general case) and one without skip connections. The optimal values of other parameters are identified through temporal cross-validation. To address potential overfitting concerns arising from the model's complexity relative to available data, the NARFIMA framework incorporates several safeguards that ensure robust generalization. The neural network architecture is deliberately constrained to a single hidden layer with a limited number of neurons ($k \leq 5$), preventing excessive parameterization while maintaining sufficient nonlinear modeling capacity. Subsequently, the NARFIMA model is trained with the optimal parameters on the entire training dataset to accurately forecast the exchange rate dynamics. The integration of ARFIMAx, which provides a robust foundation for modeling linear long-term dependencies, with the neural network, which excels in learning complex, nonlinear interactions, allows the NARFIMA architecture to capture both nonlinear and long-range dependencies. This ensures a comprehensive representation of the exchange rate series dynamics by combining long-memory dependencies, macroeconomic influences, and nonlinear fluctuations. In the next subsection, we study the asymptotic stationarity and ergodicity of the proposed NARFIMA model from a nonlinear time series perspective.

3.3. Geometric Ergodicity and Asymptotic Stationarity

Stationarity and ergodicity are fundamental for statistical inference in nonlinear time-series analysis. When a process is both stationary and ergodic, a single long realization is sufficient for time averages to recover the data-generating law. Asymptotic stationarity guarantees that distributional features stabilize as time grows, even in the presence of long memory or transient dynamics. Geometric ergodicity of the Markov chain induced by the model's state ensures exponentially fast convergence to the invariant distribution. Together, these properties justify using NARFIMA for econometric and financial forecasting under long-memory and nonlinear regimes. Building on Trapletti et al., 1999, 2000; Chakraborty et al., 2021, which analyze autoregressive neural networks (ARNN) and ARIMA-ARNN models, we develop a unified framework for NARFIMA with skip connections, a strictly more general specification. We present the theory and its economic implications and applications.

We demonstrate the asymptotic properties for the NARFIMA(1, 1, k) process with skip connections. However, for simplicity, we omit exogenous variables for establishing the theoretical results. The simple NARFIMA process is given by:

$$y_t = f(y_{t-1}, e_{t-1}, \Theta) + \varepsilon_t,$$

where Θ denotes the weight vector, ε_t is a sequence of independently and identically distributed (i.i.d.) random noise, and $f(y_{t-1}, e_{t-1}, \Theta)$ represents an autoregressive neural network with k hidden units and inputs y_{t-1} and the ARFIMA feedback e_{t-1} , as defined in Section 3.1. The output of the simple NARFIMA(1, 1, k) process with an activation function G is as follows:

$$\begin{aligned} f(y_{t-1}, e_{t-1}, \Theta) &= \psi_1 y_{t-1} + \psi_2 e_{t-1} + \beta_0 + \sum_{i=1}^k \beta_i G(\phi_{i,1} y_{t-1} + \phi_{i,2} e_{t-1} + \mu_i) \\ &\equiv \psi_1 y_{t-1} + \psi_2 e_{t-1} + g(y_{t-1}, e_{t-1}, \beta, \phi), \end{aligned} \quad (2)$$

where $\psi = (\psi_1, \psi_2)^\top$ is a weight vector representing the skip connections, the input to hidden layer weight vector is denoted by $\phi = (\phi_{1,1}, \dots, \phi_{k,1}, \phi_{1,2}, \dots, \phi_{k,2}, \mu_1, \dots, \mu_k)^\top$, and $\beta = (\beta_0, \beta_1, \dots, \beta_k)^\top$ is the hidden to output layer weight vector. Both ψ and β are incorporated into the overall weight vector Θ of the neural network. The function g represents the nonlinear transformation performed by the hidden layer, combining Θ , y_{t-1} , and e_{t-1} through the activation function G . The one-step model is given by

$$y_t = \psi_1 y_{t-1} + \psi_2 e_{t-1} + \beta_0 + \sum_{i=1}^k \beta_i G(\mu_i + \phi_{i,1} y_{t-1} + \phi_{i,2} e_{t-1}) + \varepsilon_t.$$

Now, we write the NARFIMA process in the state space form as follows:

$$x_t = \Psi x_{t-1} + F(x_{t-1}) + S e_t + \sum \varepsilon_t, \quad (3)$$

where $x_t = \begin{bmatrix} y_t \\ e_t \end{bmatrix}$, $S = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\Psi = \begin{bmatrix} \psi_1 & \psi_2 \\ 0 & 0 \end{bmatrix}$, and $F(x_{t-1}) = \begin{bmatrix} g(y_{t-1}, e_{t-1}, \beta, \phi) \\ 0 \end{bmatrix}$ is the nonlinear part. We say $\{x_t\}$ is a Markov chain with state space $\mathcal{X} \subseteq \mathcal{R}^2$ equipped with Borel σ -field $\mathcal{B}$ and Lebesgue measure λ .

In order to show the asymptotic stationarity and ergodicity of NARFIMA, we state several key assumptions: Assumption (1) is the condition for stationarity of the ARFIMA component, where the fractional differencing parameter satisfies $0 < d < \frac{1}{2}$ and AR and MA polynomials are invertible (Granger and Joyeux, 1980). This ensures ARFIMA residuals are stationary ($\{e_t\}$ is a stationary process). Assumption (2) relies on the characteristics of the activation function to be used in the neural network architecture of the NARFIMA process. Assumption (2) is essential for the stability of the dynamical system since it ensures that the behavior of the nonlinear part in Eqn. (3) is predictable and does not exhibit erratic changes. This assumption is satisfied by popularly used activation functions, such as sigmoid and tanh, and it is considered in the choice of activation function during the implementation of the NARFIMA model. Additionally, the neural network may have skip connections where output also depends linearly on inputs like previous lag and past residual. Assumptions (3) and (5) ensure contraction in the linear part and bound the influence of past states. The innovation ε_t satisfies Assumption (4), which are usual restrictions on the noise process. We verify these assumptions empirically in Section 4.5 while implementing the NARFIMA model on BRIC exchange rate datasets.

Assumption 1. Let $\{e_t\}$ be an ARFIMA($\tilde{p}, d, \tilde{q}$) process with $0 < d < \frac{1}{2}$, where the AR and MA polynomials are invertible and $\epsilon_t \sim i.i.d.$ with $\mathbb{E}(\epsilon_t) = 0$ and $\text{Var}(\epsilon_t) = \sigma^2$. Then, the residual process $\{e_t\}$ is weakly stationary with slowly decaying autocorrelation: $\rho(h) \sim ch^{2d-1}$, $h \rightarrow \infty$ for some constant $c > 0$.

Assumption 2. The activation function G is bounded, nonconstant, and asymptotically constant function. This makes the neural part F in Eqn. (3) bounded and Lipschitz.

Assumption 3. The linear part in Eqn. (3) is controllable, that is, every point of the state space can be reached irrespective of the starting point. This condition is satisfied if $\psi_1 + \psi_2 \neq 0$.

Assumption 4. The distribution of the noise process ε_t is absolutely continuous with respect to the Lebesgue measure λ with a continuous $f_\varepsilon(\cdot)$ that is strictly positive on $\mathcal{R}$. The random noise process ε_t is bounded with finite variance, i.e., $\mathbb{E}[\varepsilon_t^2] < \infty$.

Assumption 5. The skip connection weight for the AR part satisfies $|\psi_1| < 1$. For $p, q > 1$, the spectral radius of the skip companion matrix $\mathcal{M}(\Psi) < 1$ (or, equivalently, the roots of the skip AR polynomial outside the unit circle).

With the assumptions established, we now proceed to prove the asymptotic stationarity and ergodicity of the NARFIMA(1, 1, k) process. To establish geometric ergodicity and asymptotic stationarity of the NARFIMA process, we first show that the process is irreducible by proving the forward accessibility of its control system (Lemma 1 and Lemma 2). Then, we construct a suitable Lyapunov function and verify a geometric drift condition, which, together with irreducibility, ensures the existence of a unique invariant distribution to which the process converges geometrically fast, completing the proof. To start with, we initially show the irreducibility of the associated Markov chain $\{x_t\}$. A formal definition of irreducibility is provided below (Meyn and Tweedie, 2012).

Definition 1. A Markov chain is called irreducible if $\sum_{n=1}^{\infty} \mathbb{P}^n(x, \mathcal{A}) > 0$ for all $x \in \mathcal{X}$, whenever $\lambda(\mathcal{A}) > 0$, where $\mathbb{P}^n(x, \mathcal{A})$ denotes the n -step transition probability from the state x to the set $\mathcal{A} \in \mathcal{B}$, with state space $\mathcal{X} \subseteq \mathcal{R}^2$.

If for each current state x , the one-step law $\mathbb{P}(x, \cdot)$ admits a density $p(x, \cdot)$ w.r.t. Lebesgue measure that is strictly positive on every nonempty open set, then the chain is irreducible. From a control theory perspective, irreducibility is closely related to forward accessibility; for details, refer to [Meyn and Tweedie, 2012](#). Below is a formal definition of a forward accessible control system.

Definition 2. Let $A_+^t(x)$ be the set of all states which are accessible from x at time t , we set $A_+^0 := \{x\}$ and $A_+^t(x) := \{F_t(x_0, u_1, \dots, u_t); u_i \in \theta\}$, where the control set θ is an open set on the real line $\mathcal{R}$. Then, the control system F_t is said to be forward accessible if the set $\bigcup_{t=0}^{\infty} A_+^t(x)$ has nonempty interior for each $x \in \mathcal{X}$.

Since the activation function is nonconstant and bounded and $\psi_1 + \psi_2 \neq 0$ holds, the control model in Eqn. (3) is forward accessible. Hence, from any state $(y_{t-1}, e_{t-1}) = (y, e)$, the one-step reachable set contains an open interval (nonempty interior). By iterating the same argument over steps, the reachable set from any initial state has a nonempty interior. The proof of the following lemma is provided in Appendix A.1.

Lemma 1. The control system in Eqn. (3) is forward accessible provided that Assumptions (2) and (3) are satisfied.

In the following lemma, we establish the irreducibility of the Markov chain associated with the NARFIMA process. The noise has a strictly positive density, which implies irreducibility. The proof is provided in Appendix A.2.

Lemma 2. Suppose that Assumption (4) holds. Then, the Markov chain in Eqn. (3) is irreducible on the state space $(\mathcal{R}^2, \mathcal{B})$ if the conditions in Lemma 1 are satisfied.

Remark 1. A trivial example that satisfies the conditions of Lemma 2 is Gaussian white noise. Lemma 1 shows the controllability of the linear part of the NARFIMA control system, which implies forward accessibility. Although this condition is sufficient but not necessary, if the support of the noise distribution is sufficiently large, the associated Markov chain becomes irreducible.

Once irreducibility is established, we proceed to demonstrate the stationarity of the state-space formulation given in Eqn. (3). The stationarity of a Markov chain $\{x_t\}$ is closely related to the geometric ergodicity of the underlying process. Heuristically, geometric ergodicity implies that the Markov chain converges to its stationary distribution. A formal definition of geometric ergodicity and asymptotic stationarity is provided below ([Meyn and Tweedie, 2012](#)).

Definition 3. A Markov chain $\{x_t\}$ is called geometrically ergodic if there exists a probability measure Π on a probability triple $(\mathcal{X}, \mathcal{B}, \lambda)$ and a constant $\rho > 1$ such that $\lim_{n \rightarrow \infty} \rho^n \|\mathbb{P}^n(x, \cdot) - \Pi(\cdot)\| = 0$ for each $x \in \mathcal{X}$ and $\|\cdot\|$ denotes the total variation norm. Then, we say the distribution of $\{x_t\}$ converges to Π and $\{x_t\}$ is asymptotically stationary.

Building on this idea, we extend the analysis to establish the geometric ergodicity of the NARFIMA(1, 1, k) process, as formalized in the theorem below. A drift geometric condition with Lyapunov function $V(\cdot)$ is written as ([Meyn and Tweedie, 2012](#))

$$\mathbb{E}[V(y_t) \mid y_{t-1} = y, e_{t-1} = e] \leq (1 - \delta)V(y) + B \quad \text{with } \delta \in (0, 1) \text{ and } B < \infty.$$

Combining the geometric drift condition with irreducibility implies geometric ergodicity and thus asymptotic stationarity. The proof is provided in Appendix A.3.

Theorem 1. (Ergodicity) Consider the Markov chain $\{x_t\}$ associated with the NARFIMA(1, 1, k) process. Suppose that Assumptions (2) - (4) are satisfied. Then, Assumptions (1) and (5) are sufficient conditions for the geometric ergodicity of the process.

Remark 2. Theorem 1 says that, under the stated conditions (bounded activation function, contractive skip weights, stationary ARFIMA residuals, and noise process with a positive continuous density), the NARFIMA (1, 1, k) has a unique invariant law and converges to it at a geometric rate from any starting value. Practically, this implies well-defined long-run moments, reliable long-horizon forecasts (no explosive behavior), and valid ergodic averages for estimation and inference. The corollary below states that the process is strictly stationary if initialized at the invariant law.

Corollary 1. (Asymptotic Stationarity of NARFIMA Process) Let $\{y_t\}$ be the NARFIMA(1, 1, k) process satisfying the conditions of Theorem 1, then $\{y_t\}$ is asymptotically stationary.

Corollary 1 follows directly from the proof of Theorem 1, and it confirms that the NARFIMA process is asymptotically stationary, meaning its distribution stabilizes over time regardless of initialization. This allows us to use long-run statistical properties, ensures the consistency of estimators, and guarantees meaningful long-term forecasting behavior.

3.4. Economic Implications and Practical Applications

The theoretical results established for the NARFIMA(1, 1, k) process can be extended, under similar conditions, to the more general NARFIMA(p, q, k) process. Although the NARFIMA model captures long memory (frequently observed in exchange rates) dynamics, the process remains asymptotically stationary under the fractional integration parameter $0 < d < \frac{1}{2}$ and appropriate constraints on the neural component. The established results on asymptotic stationarity and geometric ergodicity of the NARFIMA model have significant practical relevance in financial and macroeconomic time series analysis. These theoretical properties justify the model's use in applied forecasting and policy modeling in the following ways:

- (a) From a practitioner's perspective, when an irreducible NARFIMA model is estimated from observed time series data, the estimated weights are likely to be close to the true underlying parameters. However, if the irreducibility conditions are not satisfied, the model may be misspecified, and any estimated weights should be interpreted with caution.
- (b) Asymptotic stationarity ensures that key statistical moments (mean, variance, and autocovariance) remain well-defined and stable over time, even in the presence of long memory. This is essential in macroeconomic and financial applications where persistent shocks such as inflation trends or volatility clusters are common.
- (c) Geometric ergodicity implies that the time series converges rapidly to its stationary distribution. This facilitates accurate inference from a single long-run realization, which is particularly useful in economics where multiple independent replications are not available.
- (d) The neural network component of NARFIMA is capable of capturing nonlinearity, asymmetry, and higher-order interactions. When combined with stationarity and ergodicity guarantees, this nonlinear structure becomes a reliable tool for modeling regime-dependent behaviors such as crisis dynamics, monetary policy shifts, and speculative bubbles.
- (e) Central banks and financial institutions often require stable, interpretable, and generalizable models. The ergodicity and asymptotic stationarity of NARFIMA ensure that its forecasts can be relied upon in real-time decision-making, particularly when designing interventions under uncertainties characterized by long memory and nonlinear dynamics.

4. Experimental Analysis

This section first analyzes the potential drivers of monthly spot exchange rates for BRIC countries by examining historical data and several global and country-specific economic indicators. We investigate the role of GEP, US EMV, US MPU, oil price growth rates, short-term interest rates, CPI inflation, and GPR index in forecasting exchange rates. This investigation allows us to identify the most relevant predictors to be used as exogenous variables in the proposed NARFIMA approach. Further, we evaluate the effectiveness of the NARFIMA model in forecasting the spot exchange rate of the BRIC economies by comparing its performance against state-of-the-art architectures from various paradigms. To assess its generalizability, we employed a rolling window forecasting approach with six different time horizons of lengths 1 month, 3 months, 6 months, 12 months, 24 months, and 48 months.

4.1. Causal Analysis

To assess the causal impact of macroeconomic covariates on the exchange rates of the BRIC economies, we employ a nonlinear Granger causality test. This choice is motivated by the global characteristics of the macroeconomic variables, which predominantly exhibit nonlinear patterns, as identified in Section 2. The nonlinear Granger causality (GC) test evaluates the temporal predictive causality, determining whether the lagged values of one variable improve the explanatory power of another variable beyond what is provided by its past observations (Granger, 1969; Hiemstra and Jones, 1994). This test specifically assesses the presence of nonlinear causal relationships between the exchange rate and macroeconomic variables. Table 2 presents the results of the nonlinear GC test across BRIC countries. As presented in the table, the p-value of the nonlinear GC test is below 0.05 for several global covariates, including GEP, US EMV, US MPU, oil price growth rate, and the country-specific short-term IRD. This indicates a significant nonlinear causation between these macroeconomic drivers and the exchange rates. The causality analysis reveals varying degrees of association between the examined variables and spot exchange rates. Notably, some indicators, such as CPI inflation and GPR, do not demonstrate strong causal relationships with exchange rates across all the BRIC economies. Based on these findings, we refine our selection of auxiliary variables for the forecasting exercise, focusing on GEP, US EMV, US MPU, oil price growth rate, and the country-specific short-term IRD that show the most consistent and significant associations with exchange rate movements.

4.2. Baseline Models

We evaluate the forecasting performance of the proposed NARFIMA model against sixteen baseline statistical and advanced deep learning models, some of which are capable of incorporating auxiliary covariates. Among the statistical frameworks, we consider *Naïve*, *Autoregressive* (AR), *Autoregressive Integrated Moving Average with exogenous variables* (ARIMAX), *Autoregressive Fractionally Integrated Moving Average with exogenous variables* (ARFIMAX), *Exponential Smoothing* (ETS), *Self-exciting Threshold Autoregressive* (SETAR), *Trigonometric Box-Cox ARIMA Trend Seasonal* (TBATS), *Generalized Autoregressive Conditional Heteroscedasticity* (GARCH), and *Bayesian Structural Time Series with exogenous variables* (BSTSx). The deep learning architectures used in this evaluation include *Autoregressive Neural Network with exogenous variables* (ARNNx), *Deep learning-based Autoregressive* (DeepAR), *Neural Basis Expansion Analysis for Time Series with exogenous variables* (NBeatsx),

Table 2. Nonlinear Granger causality test results assessing the influence of exogenous covariates on exchange rates in BRIC countries.

Exogenous Variable	Brazil exchange rates		Russia exchange rates		India exchange rates		China exchange rates	
	p-value	Conclusion	p-value	Conclusion	p-value	Conclusion	p-value	Conclusion
GEPU	0.000	Causality	0.000	Causality	0.000	Causality	0.000	Causality
US EMV	0.000	Causality	0.000	Causality	0.000	Causality	0.000	Causality
US MPU	0.000	Causality	0.000	Causality	0.000	Causality	0.000	Causality
GPR	1.000	No Causality	0.997	No Causality	1.000	No Causality	1.000	No Causality
Oil Price Growth Rate	0.000	Causality	0.000	Causality	0.000	Causality	0.000	Causality
Short-term Interest Rate	1.000	No Causality	1.000	No Causality	1.000	No Causality	1.000	No Causality
US Short-term Interest Rate	0.999	No Causality	0.975	No Causality	0.999	No Causality	0.728	No Causality
Short-term IRD	0.000	Causality	0.000	Causality	0.000	Causality	0.000	Causality
CPI Inflation	1.000	No Causality	1.000	No Causality	1.000	No Causality	1.000	No Causality
US CPI Inflation	1.000	No Causality	0.996	No Causality	1.000	No Causality	1.000	No Causality
CPI Inflation Differential	1.000	No Causality	1.000	No Causality	1.000	No Causality	1.000	No Causality

Neural Hierarchical Interpolation for Time Series with exogenous variables (NHITSx), Decomposition-based Linear model with exogenous variables (DLinearx), Normalization-based Linear model with exogenous variables (NLinearx), and Time Series Mixer with exogenous variables (TSMixerx). A detailed description of these baseline models is provided in Appendix C.1.

4.3. Evaluation Metrics

To assess the performance of the proposed NARFIMA model and the baseline frameworks in forecasting the exchange rate series of the BRIC nations, we used five popularly used evaluation metrics, namely Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), Mean Absolute Error (MAE), Mean Absolute Scaled Error (MASE), and Root Mean Square Error (RMSE). The mathematical formulations of these metrics are provided below:

$$\begin{aligned}
 \text{MAPE} &= \frac{1}{h} \sum_{t=1}^h \frac{|\hat{y}_t - y_t|}{|y_t|} \times 100\%; & \text{SMAPE} &= \frac{1}{h} \sum_{t=1}^h \frac{|\hat{y}_t - y_t|}{(|\hat{y}_t| + |y_t|)/2} \times 100\%; & \text{MAE} &= \frac{1}{h} \sum_{t=1}^h |y_t - \hat{y}_t|; \\
 \text{MASE} &= \frac{\sum_{t=1}^h |\hat{y}_t - y_t|}{\frac{h}{T-1} \sum_{t=2}^T |y_t - y_{t-1}|}; & \text{RMSE} &= \sqrt{\frac{1}{h} \sum_{t=1}^h (\hat{y}_t - y_t)^2},
 \end{aligned}$$

where y_t denotes the ground truth observation at time t with the corresponding forecast $\hat{y}_t$, T indicates the number of training observations, and h is the forecast horizon. By definition, the model with the smallest error metric value is identified as the best-performing model (Hyndman and Athanasopoulos, 2018; Panja et al., 2023).

4.4. Implementation of NARFIMA Model

This section describes the implementation and performance of the proposed NARFIMA model in forecasting the exchange rate series of the BRIC economies. To ensure a comprehensive evaluation, NARFIMA is compared with several state-of-the-art forecasting models. The NARFIMA framework is implemented in **R** statistical software using a two-stage approach. Initially, an ARFIMAx model is fitted using the *arfima* function from the ‘forecast’ package in **R**. This step captures the long-memory dependencies of the time series while incorporating significant exogenous variables, including GEPU, US EMV, US MPU, oil price growth rate, and the country-specific short-term IRD. Table 3 summarizes the estimated parameters for the ARFIMAx model used in this evaluation. In the next stage, the residuals from the ARFIMAx model, along with the training series and exogenous covariates, are modeled using a single hidden-layered feed-forward neural network. The network is built using the *nnet* function from the ‘nnet’ package in **R**, where it processes p -lagged inputs of the exchange rate series, q -lagged residuals from the ARFIMAx model, and one-lagged value from each exogenous variable through k hidden nodes to generate a one-step-ahead forecast of the target series. Multi-step-ahead forecasts are generated recursively from the NARFIMA model. To optimize model parameters, we conduct a time series-based cross-validation over the parameters (p, q, k) within the range 1 - 5, minimizing the RMSE metric. Moreover, the NARFIMA model is implemented with two variations of the feed-forward neural network: one that allows direct connections between the input and the output layers (*skip* = TRUE) and the other without the direct connections (*skip* = FALSE). The optimal parameters of the NARFIMA(p, q, k, skip) model for BRIC countries across different forecast horizons are summarized in Table 4. To implement the proposed NARFIMA model effectively, it is essential to verify the presence of nonlinearity in the residuals of the ARFIMAx model. To assess the linearity of the residuals, we employ the Terasvirta and BDS tests, which reject the null hypothesis of linearity when the computed p-value falls below 0.05 (Prabowo et al., 2020; Huang et al., 2023). The results of these tests, reported in Appendix C.2, confirm that ARFIMAx residuals exhibit significant nonlinearity. This indicates that the ARFIMAx model successfully captures the linear dependencies while leaving behind a complex nonlinear structure, which is then modeled by the neural network component of NARFIMA.

Table 3. ARFIMAx model parameters $(\bar{p}, d, \bar{q})$ for BRIC countries across 1-month-ahead ($h = 1$), 3-month-ahead ($h = 3$), 6-month-ahead ($h = 6$), 12-month-ahead ($h = 12$), 24-month-ahead ($h = 24$), and 48-month-ahead ($h = 48$) rolling window forecasts of exchange rate.

Country	$(\bar{p}, d, \bar{q})_{h=1}$	$(\bar{p}, d, \bar{q})_{h=3}$	$(\bar{p}, d, \bar{q})_{h=6}$	$(\bar{p}, d, \bar{q})_{h=12}$	$(\bar{p}, d, \bar{q})_{h=24}$	$(\bar{p}, d, \bar{q})_{h=48}$
Brazil	(0,0.493,5)	(0,0.492,5)	(0,0.492,5)	(0,0.492,5)	(0,0.493,5)	(0,0.490,5)
Russia	(0,0.490,5)	(0,0.490,5)	(0,0.489,5)	(0,0.494,2)	(0,0.489,5)	(0,0.492,3)
India	(0,0.494,5)	(0,0.493,5)	(0,0.493,5)	(0,0.493,5)	(0,0.492,5)	(0,0.491,5)
China	(0,0.496,4)	(0,0.496,4)	(0,0.496,4)	(0,0.496,4)	(0,0.497,4)	(0,0.499,1)

Table 4. Optimal NARFIMA model parameters $(p, q, k, skip)$ for BRIC countries across 1-month-ahead ($h = 1$), 3-month-ahead ($h = 3$), 6-month-ahead ($h = 6$), 12-month-ahead ($h = 12$), 24-month-ahead ($h = 24$), and 48-month-ahead ($h = 48$) rolling window forecasts of exchange rate.

Country	$(p, q, k, skip)_{h=1}$	$(p, q, k, skip)_{h=3}$	$(p, q, k, skip)_{h=6}$	$(p, q, k, skip)_{h=12}$	$(p, q, k, skip)_{h=24}$	$(p, q, k, skip)_{h=48}$
Brazil	(5,4,4,F)	(2,5,2,F)	(1,2,1,F)	(1,1,1,F)	(2,5,1,F)	(4,2,1,T)
Russia	(1,2,2,T)	(1,1,1,F)	(1,1,1,F)	(5,2,5,T)	(1,1,5,T)	(3,2,1,F)
India	(2,1,1,T)	(4,3,4,T)	(4,2,1,T)	(1,3,4,T)	(5,4,1,T)	(2,4,4,T)
China	(5,4,1,T)	(1,2,1,T)	(1,1,2,F)	(5,4,1,T)	(1,2,4,T)	(4,1,2,T)

4.5. Validation of Theoretical Results on Empirical Data

We empirically investigate the validity of the theoretical assumptions underlying the NARFIMA model. Assumption (2) is ensured by the model design, since the hidden layer employs a logistic (sigmoid) activation function, defined as $\sigma(x) = \frac{1}{1+e^{-x}}$. The function is (i) bounded in $(0, 1)$; (ii) $\lim_{x \rightarrow -\infty} \sigma(x) = 0$, $\lim_{x \rightarrow +\infty} \sigma(x) = 1$ (asymptotically constant); (iii) $\sigma'(x) = \sigma(x)(1-\sigma(x)) \leq \frac{1}{4}$ implies $1/4$ -Lipschitz and $\sigma_\alpha(x) = \sigma(\alpha x)$ is $\alpha/4$ -Lipschitz. These properties ensure that Assumption (2) is satisfied in the NARFIMA implementation. We further evaluate Assumptions (3) and (5) using BRIC exchange rate data, extending the theoretical framework to the NARFIMA($p, q, k, skip$) process. In this setting, Assumption (3) generalizes to the condition $\sum_{i=1}^p \psi_{1,i} + \sum_{j=1}^q \psi_{2,j} \neq 0$, where $\psi_{1,i}$ and $\psi_{2,j}$ denote the autoregressive and residual skip weights, respectively, while Assumption (5) requires $|\sum_{i=1}^p \psi_{1,i}| < 1$. Table 5 reports the results for the long forecast horizons of 12, 24, and 48 months, for which the optimal parameter configurations in Table 4 had skip connections ($skip = \text{TRUE}$). Empirically, both Assumptions (3) and (5) were satisfied across all countries, confirming the validity of the theoretical framework in practice.

Table 5. Empirical validation of Assumptions (3) and (5) for the NARFIMA model. Long horizons with $skip = \text{FALSE}$ are omitted, as skip connections were removed.

Country	Horizon	NARFIMA($p, q, k, skip$)	AR Skip Weights	Residual Skip Weights	Assumption (3)	Assumption (5)
Brazil	48	(4,2,1,T)	(-0.456,1.249,-0.452,0.326)	(-0.176,0.015)	0.506	0.667
Russia	12	(5,2,5,T)	(-1.055,0.057,-0.129,0.096,0.273)	(0.150,-0.069)	-0.677	0.758
	24	(1,1,5,T)	-0.097	0.694	0.597	0.097
India	12	(1,3,4,T)	-0.974	(0.965,0.115,0.241)	0.347	0.974
	24	(5,4,1,T)	(-0.497,-0.021,-0.049,0.387,0.119)	(0.338,-0.076,-0.234,-0.012)	-0.045	0.061
	48	(2,4,4,T)	(-0.124,0.346)	(0.756,-0.029,-0.187,0.043)	0.805	0.222
China	12	(5,4,1,T)	(-0.338,0.380,0.172,0.310,-0.152)	(0.213,0.029,-0.044,-0.011)	0.559	0.372
	24	(1,2,4,T)	0.464	(0.876,0.011)	1.351	0.464
	48	(4,1,2,T)	(-0.547,1.217,-0.954,0.962)	-0.445	0.233	0.678

4.6. Baseline Comparisons

After implementing the proposed NARFIMA model, we implemented the baseline forecasters and generated exchange rate forecasts of BRIC economies for multiple time horizons. Among the baseline forecasters, the classical time series models ARIMAx, ARFIMAx, ETS, TBATS, and ARNNx are implemented using the ‘forecast’ package in **R**. The implementation of the Naïve, AR, SETAR, BSTSx, and GARCH models is adopted from ‘stats’, ‘tsDyn’, ‘bsts’, and ‘tseries’ packages in **R**, respectively. Additionally, the deep learning models DeepAR, NBeatsx, NHiTSx, DLinearx, NLinearx, and TSMixerx are implemented using the ‘darts’ library in Python. The forecasting performance for Brazil, Russia, India, and China is presented in Tables 6, 7, 8, and 9, respectively. These tables highlight that for most forecasting tasks, the NARFIMA model demonstrates superior performance over the baseline architectures. For short-term forecasting (1-month-ahead horizon), NARFIMA significantly outperforms all baseline models across the BRIC economies, as indicated by all performance metrics. For the 3-month-ahead forecasting horizon, NARFIMA provides the most accurate forecasts for Brazil and China, while ARIMAx performs competitively for Russia and India. In the semi-long-term 6-month-ahead forecasts, the NARFIMA model substantially reduces forecast errors compared to its component frameworks, ARFIMAx and ARNNx, across all BRIC nations and provides the best exchange rate forecasts for India and China. Among the baseline models, ARIMAx and ETS produce comparable forecasts for Brazil and Russia. In the case of 12-month-ahead forecasts of BRIC exchange rates, the NARFIMA model generates the most accurate forecasts for Brazil and Russia, while the Naïve and ARIMAx framework performs better for India and China, respectively. For long-term forecasting (24-month-ahead horizon), the NARFIMA model maintains its performance supremacy across all BRIC countries except India, where NLinearx generates more accurate forecasts. At the 48-month-ahead forecasts, our proposed NARFIMA model consistently outperforms all competing

frameworks, highlighting its ability to capture both long-term dependencies and nonlinear patterns in time series data. From a country-specific perspective, NARFIMA delivers the most accurate forecasts in five out of six horizons for Brazil and China. For Russia, it outperforms baseline models in four out of six cases while remaining competitive with ARIMAx and ETS models in the other two horizons. However, for India's exchange rate series, the NARFIMA model performs best in only three forecasting horizons. This is primarily attributed to the linearity of India's exchange rate data, which is better modeled with ARIMAx, Naïve, and NLinearx frameworks. These empirical results confirm that the NARFIMA approach can effectively model volatility, non-stationarity, and nonlinearity in time series data. However, its forecasting performance declines for inherently linear series, where traditional statistical models perform better. Nevertheless, as most macroeconomic variables exhibit nonlinear characteristics (Franses and Van Dijk, 2000), NARFIMA remains well-suited for forecasting complex financial time series. On the other hand, deep learning models yield relatively inaccurate exchange rate forecasts compared to both statistical baselines and the NARFIMA model. This is likely due to the low sample size of the dataset, which pose significant challenges in accurately training deep learning architectures. While some baseline models, including ARIMAx and ETS, achieve comparable short-term and semi-long-term performance, their accuracy deteriorates significantly over longer horizons. In contrast, NARFIMA consistently performs well across all time horizons, demonstrating its robustness and generalizability. Additionally, the integration of ARFIMAx feedback residuals with the target series and exogenous variables in NARFIMA proves to be a highly effective technique for capturing long-term dependencies, especially for non-stationary and nonlinear exchange rate series.

Table 6. Evaluation of the proposed NARFIMA model's performance relative to baseline forecasters across all forecast horizons for Brazil (**best** and **second-best** results are highlighted). The MASE metric is not defined for a forecast horizon of length one.

Horizon	Metrics	Naïve	AR	ARIMAx	ARNNx	ARFIMAx	ETS	SETAR	TBATS	GARCH	BSTsX	DeepAR	NBeatsx	NHiTSx	DLinearx	NLinearx	TSMixerx	NARFIMA
1	MAPE	2.318	2.855	5.613	5.446	2.270	1.949	2.084	2.351	2.350	5.965	1.812	8.493	9.267	11.001	16.562	22.367	0.041
	SMAPE	2.345	2.896	5.775	5.599	2.296	1.969	2.106	2.379	2.378	6.148	1.829	8.869	8.857	10.428	15.296	25.183	0.042
	MAE	0.117	0.144	0.284	0.275	0.115	0.099	0.105	0.119	0.119	0.302	0.092	0.429	0.469	0.556	0.837	1.131	0.002
	RMSE	0.117	0.144	0.284	0.275	0.115	0.099	0.105	0.119	0.119	0.302	0.092	0.429	0.469	0.556	0.837	1.131	0.002
	MASE	3.301	4.688	6.368	9.182	4.995	5.137	2.603	3.254	3.304	6.789	0.761	2.512	1.284	17.789	17.603	13.737	0.541
3	MAPE	3.365	4.818	6.585	9.698	5.142	5.291	2.642	3.316	3.368	7.033	0.761	2.478	1.276	16.324	16.170	14.757	0.541
	SMAPE	3.365	4.818	6.585	9.698	5.142	5.291	2.642	3.316	3.368	7.033	0.761	2.478	1.276	16.324	16.170	14.757	0.541
	MAE	0.165	0.234	0.317	0.458	0.249	0.256	0.130	0.162	0.165	0.338	0.038	0.125	0.063	0.882	0.875	0.682	0.027
	MASE	2.130	3.025	4.098	5.924	3.222	3.313	1.679	2.100	2.132	4.367	0.490	1.612	0.820	11.409	11.310	8.812	0.348
	RMSE	0.177	0.251	0.324	0.494	0.267	0.274	0.139	0.175	0.178	0.342	0.040	0.153	0.068	0.886	0.879	0.683	0.031
6	MAPE	2.280	2.263	1.265	7.136	4.903	2.198	2.503	1.444	2.251	2.239	3.841	16.281	4.841	15.575	10.943	14.721	1.370
	SMAPE	2.245	2.300	1.271	7.536	5.075	2.235	2.458	1.438	2.217	2.269	3.754	15.635	4.690	14.377	10.213	16.011	1.368
	MAE	0.111	0.112	0.063	0.354	0.243	0.109	0.122	0.071	0.110	0.111	0.188	0.800	0.238	0.772	0.540	0.725	0.067
	MASE	1.273	1.288	0.718	4.055	2.785	1.252	1.396	0.812	1.257	1.272	2.150	9.169	2.723	8.849	6.195	8.307	0.772
	RMSE	0.130	0.146	0.075	0.432	0.286	0.142	0.148	0.085	0.128	0.122	0.206	0.929	0.275	0.891	0.628	0.756	0.082
12	MAPE	4.069	2.328	3.853	4.050	5.784	5.259	4.978	3.482	3.939	3.374	3.366	25.537	6.992	14.187	15.544	26.956	1.548
	SMAPE	3.941	2.373	3.822	4.163	5.991	5.064	4.803	3.386	3.818	3.446	3.441	23.427	7.150	12.946	13.967	31.741	1.562
	MAE	0.201	0.118	0.194	0.206	0.291	0.260	0.246	0.172	0.194	0.173	0.173	1.284	0.356	0.705	0.771	1.373	0.079
	MASE	2.837	1.669	2.737	2.907	4.114	3.677	3.480	2.426	2.746	2.450	2.441	18.139	5.031	9.963	10.893	19.407	1.118
	RMSE	0.252	0.150	0.207	0.238	0.317	0.313	0.296	0.220	0.245	0.212	0.199	1.477	0.402	0.873	0.950	1.453	0.096
24	MAPE	8.246	7.324	10.068	19.586	10.250	18.685	13.438	10.277	7.938	9.671	15.602	20.290	18.297	13.855	7.850	26.467	3.341
	SMAPE	7.836	7.632	9.477	22.621	10.989	16.875	12.395	9.673	7.555	9.131	16.996	19.500	18.455	14.425	8.487	32.126	3.361
	MAE	0.414	0.374	0.507	1.002	0.524	0.944	0.676	0.517	0.398	0.487	0.805	1.029	0.939	0.729	0.417	1.382	0.173
	MASE	3.480	3.140	4.261	8.422	4.405	7.938	5.683	4.347	3.349	4.091	6.766	8.653	7.897	6.129	3.502	11.619	1.455
	RMSE	0.464	0.417	0.566	1.160	0.601	1.007	0.754	0.566	0.447	0.540	0.827	1.276	1.079	0.905	0.585	1.594	0.204
48	MAPE	20.214	36.403	17.081	45.771	34.701	18.714	28.760	20.465	20.309	15.632	36.783	43.228	32.738	25.167	26.632	31.646	11.888
	SMAPE	22.753	45.436	18.914	61.606	42.697	20.895	34.036	23.063	22.871	17.239	45.540	34.096	27.344	22.149	23.322	38.054	12.856
	MAE	1.064	1.901	0.900	2.382	1.815	0.986	1.507	1.077	1.069	0.827	1.919	2.198	1.662	1.279	1.352	1.641	0.630
	MASE	6.700	11.969	5.669	15.000	11.431	6.210	9.494	6.780	6.730	5.210	12.083	13.841	10.470	8.055	8.512	10.332	3.968
	RMSE	1.126	1.987	0.968	2.508	1.891	1.049	1.574	1.138	1.131	0.912	1.974	2.475	1.833	1.322	1.383	1.700	0.715

4.7. Robustness and Statistical Significance Tests

In this section, we assess the robustness of our empirical results by evaluating the performance of different forecasting models based on differences in measurement errors, using multiple comparisons with the best (MCB) test. The model-agnostic MCB test is a nonparametric method that ranks each forecaster based on its performance across multiple datasets (Koning et al., 2005). It then identifies the model with the lowest average rank as the best-performing framework and considers the critical distance (CD) of this model as the reference value for comparison. Fig. 1 presents the MCB test results across different forecasting tasks based on RMSE, MAPE, SMAPE, and MAE metrics. The figure shows that the proposed NARFIMA model achieves the lowest average ranks for RMSE, MAPE, SMAPE, and MAE metrics, making it the 'best' performing model, followed by the ARIMAx, BSTsX, and Naïve frameworks in terms of the RMSE metric. The CD of the NARFIMA model (shaded area), serves as the reference value of the MCB test. Since the CD values of all baseline forecasters, except ARIMAx, BSTsX, and Naïve, lie well beyond this reference, their performance differs significantly from that of the best-performing NARFIMA model. Overall, the MCB test highlights the statistical significance of the performance differences, demonstrating the superiority of the NARFIMA model across various datasets and forecast horizons.

Alongside the MCB test, we utilize the Murphy diagram approach to assess the robustness and forecastability of the proposed NARFIMA model. The Murphy diagram technique detects empirical forecast dominance between competing models across a range of scoring functions (Ehm et al., 2016). Unlike the MCB method, which focuses on ranking models based on their average forecast accuracy, the Murphy diagram provides a comprehensive evaluation by examining whether a forecasting model consistently

Table 7. Evaluation of the proposed NARFIMA model's performance relative to baseline forecasters across all forecast horizons for Russia ('best' and 'second-best' results are highlighted). The MASE metric is not defined for a forecast horizon of length one.

Horizon	Metrics	Naïve	AR	ARIMAX	ARNNx	ARFIMAX	ETS	SETAR	TBATS	GARCH	BSTsX	DeepAR	NBeatsx	NHiTSx	DLinearx	NLinearx	TSMixerx	NARFIMA
1	MAPE	4.473	3.028	5.213	12.051	2.891	6.237	4.548	4.604	1.193	5.164	74.878	25.264	18.440	3.997	10.679	40.493	0.043
	SMAPE	4.376	2.983	5.080	12.824	2.849	6.048	4.447	4.500	1.186	5.034	119.689	28.917	20.312	4.079	11.281	50.772	0.043
	MAE	4.171	2.824	4.860	11.237	2.695	5.816	4.241	4.293	1.113	4.815	69.819	23.557	17.194	3.727	9.958	37.757	0.040
	RMSE	4.171	2.824	4.860	11.237	2.695	5.816	4.241	4.293	1.113	4.815	69.819	23.557	17.194	3.727	9.958	37.757	0.040
3	MAPE	4.732	7.064	2.576	27.106	6.487	2.830	9.698	4.224	8.972	2.932	76.133	25.671	23.114	6.731	11.367	53.993	3.181
	SMAPE	4.863	7.332	2.627	31.593	6.716	2.784	10.221	4.331	9.409	2.996	122.931	29.515	26.169	7.027	12.092	73.996	3.258
	MAE	4.551	6.767	2.493	25.858	6.197	2.668	9.260	4.065	8.599	2.835	72.736	24.573	22.118	6.481	10.897	51.599	3.076
	MASE	1.609	2.392	0.881	9.141	2.191	0.943	3.273	1.437	3.040	1.002	25.713	8.687	7.819	2.291	3.852	18.240	1.088
	RMSE	4.867	6.901	3.059	26.369	6.354	3.284	9.506	4.416	8.768	3.421	72.757	24.779	22.255	7.260	11.205	51.651	3.762
6	MAPE	11.051	21.548	11.707	38.045	16.097	5.501	18.408	8.867	14.591	11.098	74.090	16.626	11.537	9.177	16.571	59.835	8.797
	SMAPE	11.893	24.438	12.636	47.493	18.004	5.683	20.701	9.389	15.932	11.908	117.747	18.149	12.311	9.696	18.339	85.465	9.306
	MAE	10.372	19.947	10.971	34.973	15.123	5.130	17.186	8.311	13.577	10.378	67.432	15.134	10.656	8.121	14.794	54.391	8.241
	MASE	2.481	4.772	2.625	8.367	3.618	1.227	4.112	1.988	3.248	2.483	16.133	3.621	2.549	1.943	3.539	13.013	1.972
	RMSE	11.817	21.085	12.385	35.989	17.433	5.806	18.987	9.421	14.724	11.580	67.671	15.257	11.209	8.970	15.700	54.556	9.297
12	MAPE	23.259	25.521	25.434	22.431	22.705	14.957	19.294	22.497	24.575	21.056	72.813	19.963	22.332	9.857	9.176	44.246	7.201
	SMAPE	27.127	30.280	29.862	25.967	26.535	16.540	21.942	26.105	28.858	24.238	114.790	17.722	19.852	10.655	8.595	56.883	7.539
	MAE	20.152	22.122	21.827	19.391	19.776	13.055	16.780	19.502	21.236	18.294	59.849	15.490	17.499	8.265	7.194	36.201	6.202
	MASE	5.278	5.794	5.716	5.078	5.179	3.419	4.395	5.108	5.562	4.791	15.674	4.057	4.583	2.165	1.884	9.481	1.624
	RMSE	22.908	25.172	24.205	21.911	23.012	15.353	19.374	22.228	23.954	21.016	60.862	16.750	17.939	10.523	8.553	36.650	7.613
24	MAPE	14.531	17.836	13.614	15.995	18.086	14.531	14.509	14.779	14.480	13.908	69.848	21.984	16.352	36.526	39.446	19.307	7.913
	SMAPE	14.905	20.107	13.064	16.787	20.722	14.905	14.838	15.384	14.642	13.860	107.824	19.221	14.614	29.079	30.997	22.362	7.722
	MAE	11.044	14.493	9.652	12.329	14.903	11.044	10.996	11.383	10.856	10.276	53.031	15.592	11.020	24.952	27.328	15.234	5.347
	MASE	2.441	3.204	2.134	2.725	3.295	2.441	2.431	2.516	2.272	11.723	3.447	2.436	5.516	6.041	3.368	3.368	1.182
	RMSE	13.305	18.504	11.396	14.174	19.165	13.305	13.311	13.730	13.053	12.064	54.491	17.803	13.578	28.648	31.076	18.372	7.488
48	MAPE	14.478	27.687	11.599	30.146	27.016	14.478	13.559	14.462	14.416	11.321	73.245	66.489	44.946	29.673	40.079	29.388	9.371
	SMAPE	15.866	33.438	12.120	38.900	32.244	15.866	14.684	15.846	15.805	11.600	115.810	46.700	35.699	23.658	30.969	35.623	9.383
	MAE	11.324	21.460	8.793	23.286	20.931	11.324	10.553	11.311	11.286	8.434	54.370	47.375	31.682	20.691	28.248	22.386	6.904
	MASE	3.268	6.193	2.537	6.720	6.040	3.268	3.045	3.264	3.257	2.434	15.690	13.671	9.142	5.971	8.152	6.460	1.992
	RMSE	13.730	24.704	10.520	29.103	23.667	13.730	12.836	13.715	13.747	10.021	55.195	53.181	33.387	26.450	33.566	24.753	9.709

Table 8. Evaluation of the proposed NARFIMA model's performance relative to baseline forecasters across all forecast horizons for India ('best' and 'second-best' results are highlighted). The MASE metric is not defined for a forecast horizon of length one.

Horizon	Metrics	Naïve	AR	ARIMAX	ARNNx	ARFIMAX	ETS	SETAR	TBATS	GARCH	BSTsX	DeepAR	NBeatsx	NHiTSx	DLinearx	NLinearx	TSMixerx	NARFIMA
1	MAPE	0.176	0.614	0.773	3.866	0.774	0.022	0.013	0.155	0.394	0.987	69.916	1.964	0.577	4.459	3.082	18.362	0.006
	SMAPE	0.176	0.616	0.770	3.942	0.777	0.022	0.013	0.156	0.395	0.992	107.494	1.945	0.578	4.561	3.130	20.218	0.006
	MAE	0.146	0.511	0.643	3.217	0.644	0.018	0.011	0.129	0.328	0.821	58.182	1.634	0.480	3.711	2.565	15.280	0.005
	RMSE	0.146	0.511	0.643	3.217	0.644	0.018	0.011	0.129	0.328	0.821	58.182	1.634	0.480	3.711	2.565	15.280	0.005
3	MAPE	1.055	1.899	0.133	7.131	3.087	1.063	0.830	1.060	1.331	0.428	70.055	2.394	5.507	4.102	7.598	18.409	0.546
	SMAPE	1.061	1.918	0.133	7.421	3.142	1.069	0.834	1.066	1.340	0.429	107.823	2.360	5.337	4.191	7.912	20.279	0.548
	MAE	0.876	1.577	0.110	5.925	2.565	0.883	0.690	0.880	1.106	0.355	58.169	1.986	4.576	3.407	6.311	15.286	0.454
	MASE	4.338	7.808	0.547	29.330	12.697	4.371	3.414	4.358	5.474	1.758	287.964	9.833	22.653	16.867	31.243	75.674	2.246
	RMSE	0.892	1.640	0.121	6.199	2.745	0.899	0.695	0.896	1.135	0.355	58.169	2.178	4.923	3.473	6.449	15.298	0.477
6	MAPE	0.798	2.324	0.578	8.584	4.733	0.892	0.436	0.828	1.235	0.329	70.874	9.691	3.016	3.224	4.742	18.645	0.176
	SMAPE	0.802	2.359	0.575	9.024	4.886	0.898	0.437	0.833	1.245	0.328	109.777	9.223	2.966	3.306	4.860	20.565	0.176
	MAE	0.661	1.925	0.476	7.104	3.921	0.740	0.361	0.686	1.023	0.271	58.564	8.016	2.493	2.654	3.916	15.408	0.145
	MASE	2.753	8.016	1.980	29.578	16.326	3.079	1.503	2.857	4.260	1.130	243.834	33.373	10.382	11.052	16.305	64.151	0.603
	RMSE	0.784	2.170	0.593	7.571	4.517	0.864	0.428	0.806	1.186	0.314	58.565	8.216	2.645	3.294	3.964	15.418	0.236
12	MAPE	0.447	3.333	0.535	9.260	6.401	2.511	1.544	0.712	1.470	1.559	72.896	14.840	4.956	2.240	4.377	19.576	1.296
	SMAPE	0.448	3.411	0.533	9.762	6.705	2.477	1.531	0.709	1.485	1.546	114.703	13.786	4.774	2.226	4.496	21.721	1.296
	MAE	0.369	2.754	0.439	7.637	5.286	2.068	1.272	0.585	1.215	1.285	60.049	12.228	4.093	1.846	3.599	16.122	1.068
	MASE	0.894	6.663	1.063	18.481	12.790	5.004	3.078	1.415	2.939	3.110	145.311	29.590	9.906	4.467	8.708	39.014	2.584
	RMSE	0.471	3.219	0.533	8.040	6.295	2.142	1.336	0.677	1.434	1.360	60.051	12.432	5.115	2.231	3.946	16.184	1.160
24	MAPE	6.176	10.081	4.962	10.541	12.650	5.159	5.831	5.729	8.251	4.212	72.293	17.215	10.885	5.669	3.275	19.947	4.555
	SMAPE	6.440	10.795	5.125	11.224	13.770	5.344	6.065	5.955	8.724	4.329	113.241	15.797	10.267	5.448	3.226	22.176	4.671
	MAE	5.041	8.222	4.043	8.539	10.308	4.213	4.759	4.676	6.731	3.432	57.782	13.743	8.648	4.553	2.629	15.981	3.637
	MASE	7.652	12.482	6.138	12.963	15.648	6.395	7.224	7.098	10.217	5.210	87.715	20.862	13.128	6.912	3.992	24.260	5.520
	RMSE	5.827	9.454	4.610	9.221	11.732	4.915	5.505	5.423	7.745	3.932	57.863	14.059	9.071	5.475	3.129	16.093	3.801
48	MAPE	7.308	16.106	4.749	18.908	17.576	6.775	8.633	7.311	11.368	3.943	72.878	9.460	26.766	3.282	3.610	31.710	2.872
	SMAPE	7.704	17.952	4.926	21.937	19.594	7.130	9.165	7.707	12.302	4.070	114.694	8.930	23.570	3.315	3.582	37.715	2.873
	MAE	5.791	12.674	3.772	14.980	13.756	5.381	6.823	5.793	8.976	3.137	56.022	7.166	20.475	2.504	2.764	24.400	2.210 </

Table 9. Evaluation of the proposed NARFIMA model’s performance relative to baseline forecasters across all forecast horizons for China (‘best’ and ‘second-best’ results are highlighted). The MASE metric is not defined for a forecast horizon of length one.

Horizon	Metrics	Naïve	AR	ARIMAx	ARNNx	ARFIMAx	ETS	SETAR	TBATS	GARCH	BSTsX	DeepAR	NBeatsx	NHiTSx	DLinearx	NLinearx	TSMixerx	NARFIMA
1	MAPE	0.126	0.018	2.588	0.487	0.048	0.381	0.149	0.262	0.349	0.202	0.874	66.993	45.806	15.326	11.975	6.737	0.005
	SMAPE	0.126	0.018	2.622	0.486	0.048	0.380	0.149	0.262	0.349	0.202	0.878	50.183	37.270	14.235	11.299	6.972	0.005
	MAE	0.009	0.001	0.189	0.036	0.003	0.028	0.011	0.019	0.025	0.015	0.064	4.895	3.347	1.120	0.875	0.492	3.8×10^{-4}
	RMSE	0.009	0.001	0.018	0.028	0.036	0.028	0.011	0.019	0.025	0.015	0.064	4.895	3.347	1.120	0.875	0.492	3.8×10^{-4}
3	MAPE	1.347	1.249	3.645	0.404	3.067	0.310	4.195	1.727	1.551	1.552	0.616	51.420	54.301	19.451	12.578	6.096	0.142
	SMAPE	1.357	1.257	3.713	0.403	3.125	0.310	4.300	1.743	1.563	1.565	0.618	40.470	42.692	17.725	11.834	6.288	0.142
	MAE	0.098	0.091	0.266	0.030	0.224	0.023	0.306	0.126	0.113	0.113	0.045	3.746	3.955	1.417	0.916	0.444	0.010
	MASE	3.358	3.113	9.080	1.009	7.651	0.771	10.462	4.305	3.865	3.868	1.536	128.061	135.213	48.435	31.323	15.184	0.354
	RMSE	0.102	0.094	0.267	0.044	0.245	0.024	0.330	0.130	0.116	0.114	0.051	3.868	3.958	1.417	0.916	0.445	0.011
6	MAPE	4.287	4.177	6.301	3.028	5.704	4.105	4.066	4.631	4.258	3.842	1.648	72.662	53.540	18.032	13.035	7.543	0.445
	SMAPE	4.393	4.277	6.515	3.078	5.885	4.202	4.160	4.753	4.362	3.929	1.662	53.191	42.181	16.533	12.229	7.852	0.445
	MAE	0.310	0.302	0.455	0.219	0.412	0.297	0.294	0.335	0.308	0.278	0.119	5.226	3.849	1.296	0.937	0.544	0.032
	MASE	4.821	4.697	7.071	3.399	6.407	4.616	4.571	5.206	4.789	4.323	1.852	81.232	59.819	20.150	14.560	8.462	0.495
	RMSE	0.329	0.320	0.466	0.227	0.429	0.315	0.310	0.353	0.327	0.299	0.127	5.250	3.859	1.299	0.941	0.557	0.039
12	MAPE	2.525	2.849	2.104	10.482	3.104	7.759	3.991	6.032	2.417	5.424	3.393	65.403	62.941	37.394	21.503	6.537	2.561
	SMAPE	2.480	2.786	2.100	9.916	3.107	7.440	3.872	5.828	2.386	5.253	3.305	49.002	47.794	31.496	19.393	6.798	2.515
	MAE	0.175	0.197	0.148	0.739	0.220	0.545	0.277	0.422	0.168	0.379	0.235	4.612	4.437	2.639	1.514	0.466	0.178
	MASE	2.108	2.374	1.783	8.891	2.642	6.553	3.328	5.080	2.026	4.560	2.828	55.495	53.396	31.758	18.225	5.613	2.142
	RMSE	0.221	0.250	0.158	0.771	0.243	0.570	0.341	0.452	0.200	0.413	0.294	4.678	4.451	2.641	1.521	0.508	0.212
24	MAPE	6.318	5.429	6.708	4.734	3.510	7.290	5.386	6.909	6.226	8.356	7.349	40.017	53.031	23.797	23.402	5.306	1.782
	SMAPE	6.608	5.639	7.042	4.896	3.589	7.686	5.597	7.264	5.931	8.864	6.971	33.087	41.582	21.217	20.921	5.467	1.780
	MAE	0.445	0.382	0.473	0.333	0.246	0.514	0.379	0.487	0.410	0.588	0.486	2.727	3.596	1.628	1.598	0.370	0.122
	MASE	5.613	4.820	5.963	4.200	3.105	6.481	4.786	6.142	5.175	7.415	6.133	34.410	45.376	20.547	20.171	4.675	1.535
	RMSE	0.531	0.454	0.566	0.397	0.300	0.615	0.454	0.582	0.513	0.693	0.583	2.797	3.656	1.651	1.610	0.427	0.150
48	MAPE	5.337	5.529	4.571	14.801	4.306	5.800	3.698	5.903	5.339	4.754	8.877	89.715	119.670	68.479	72.332	5.688	2.476
	SMAPE	5.130	5.301	4.427	13.639	4.184	5.555	3.710	5.649	5.132	4.615	8.399	61.242	74.691	50.910	53.004	5.944	2.480
	MAE	0.351	0.363	0.301	0.990	0.284	0.381	0.251	0.388	0.351	0.314	0.588	6.067	8.084	4.627	4.888	0.398	0.169
	MASE	5.420	5.609	4.652	15.309	4.390	5.889	3.883	5.993	5.423	4.856	9.086	93.786	124.956	71.519	75.559	6.147	2.610
	RMSE	0.432	0.450	0.375	1.056	0.349	0.468	0.298	0.476	0.433	0.384	0.659	6.217	8.100	4.638	4.902	0.494	0.199

different models across a range of θ values. The parameter θ , plays a crucial role in evaluating forecast accuracy, as different values emphasize different aspects of prediction error. Smaller values of θ make the scoring function more sensitive to underpredictions, penalizing models that consistently underestimate actual values, whereas higher values of θ penalize overpredictions more heavily. This adaptability makes the Murphy diagram a robust tool for comprehensive model evaluation, as it provides insights into how forecasting errors are distributed and whether a model consistently outperforms its competitors under different scoring functions.

To empirically validate the effectiveness of NARFIMA against benchmark models, we utilized the ‘murphydiagram’ package in **R** to generate the Murphy diagrams. Fig. 2 presents the Murphy diagrams comparing NARFIMA with the top-performing ARIMAx and BSTSx frameworks, as identified by the RMSE-based MCB test results, for each BRIC nation over a 48-month-ahead forecast horizon. The diagram plots the extremal scores for competing models, where a lower score indicates better model performance. The results reveal distinct patterns across different economies. In the case of Brazil, NARFIMA consistently outperforms both ARIMAx and BSTSx across all scoring functions, establishing its superiority in exchange rate forecasting. For Russia, the model provides similar or more accurate exchange rate forecasts when θ lies below 72 or above 75, suggesting that its dominance depends on specific error considerations. In India and China, NARFIMA remains competitive; however, sometimes ARIMAx and BSTSx outperform it. Overall, these findings establish NARFIMA as an accurate and reliable forecasting model across diverse economic conditions. The combination of the MCB test and the Murphy diagram provides a robust validation of NARFIMA’s forecasting superiority, confirming that it consistently outperforms benchmark methods across various scoring functions and forecast horizons.

5. Uncertainty Quantification and Ablation Study

5.1. Conformal Prediction

Alongside the point forecasts of the exchange rate dynamics, we quantify the uncertainty associated with the NARFIMA model predictions using conformal prediction intervals. The distribution-free conformal prediction approach converts uncertainty scores to prediction intervals that contain the true outcome (Vovk et al., 2005). This model-agnostic method offers several advantages over simulation-based prediction intervals, including computational efficiency, fewer assumptions about the underlying data distribution, and guarantees coverage. In the context of time series setup, the conformal prediction framework utilizes the sequential ordering of the data to generate the prediction interval. Given the training set $\{(y_t, \tilde{x}_t)_{t=1}^T\}$, where y_t represents the target series and $\tilde{x}_t$ indicates the set of features including lagged values of y_t , ARFIMAx residuals, and the exogenous variables, we apply the NARFIMA framework and an uncertainty model ($\hat{\Psi}$) on $\tilde{x}_t$ to generate a measure of uncertainty. The conformal score $\tilde{S}_t$ is then calculated as: $\tilde{S}_t = \frac{|y_t - \text{NARFIMA}(\tilde{x}_t)|}{\hat{\Psi}(\tilde{x}_t)}$. Due to the temporal dependencies in the series, the conformal quantiles, computed using a weighted conformal method with a fixed window of size τ defined as $\omega_{\tilde{t}} = \mathbb{1}\{\tilde{t} \geq t - \tau\} \quad \forall \tilde{t} < t$, are as follows:

$$\text{CQ}_t = \inf \left\{ \tilde{q} : \frac{1}{\min(\tau, \tilde{t} - 1) + 1} \sum_{\tilde{t}=1}^{t-1} \tilde{S}_{\tilde{t}} \omega_{\tilde{t}} \geq 1 - \alpha \right\}.$$

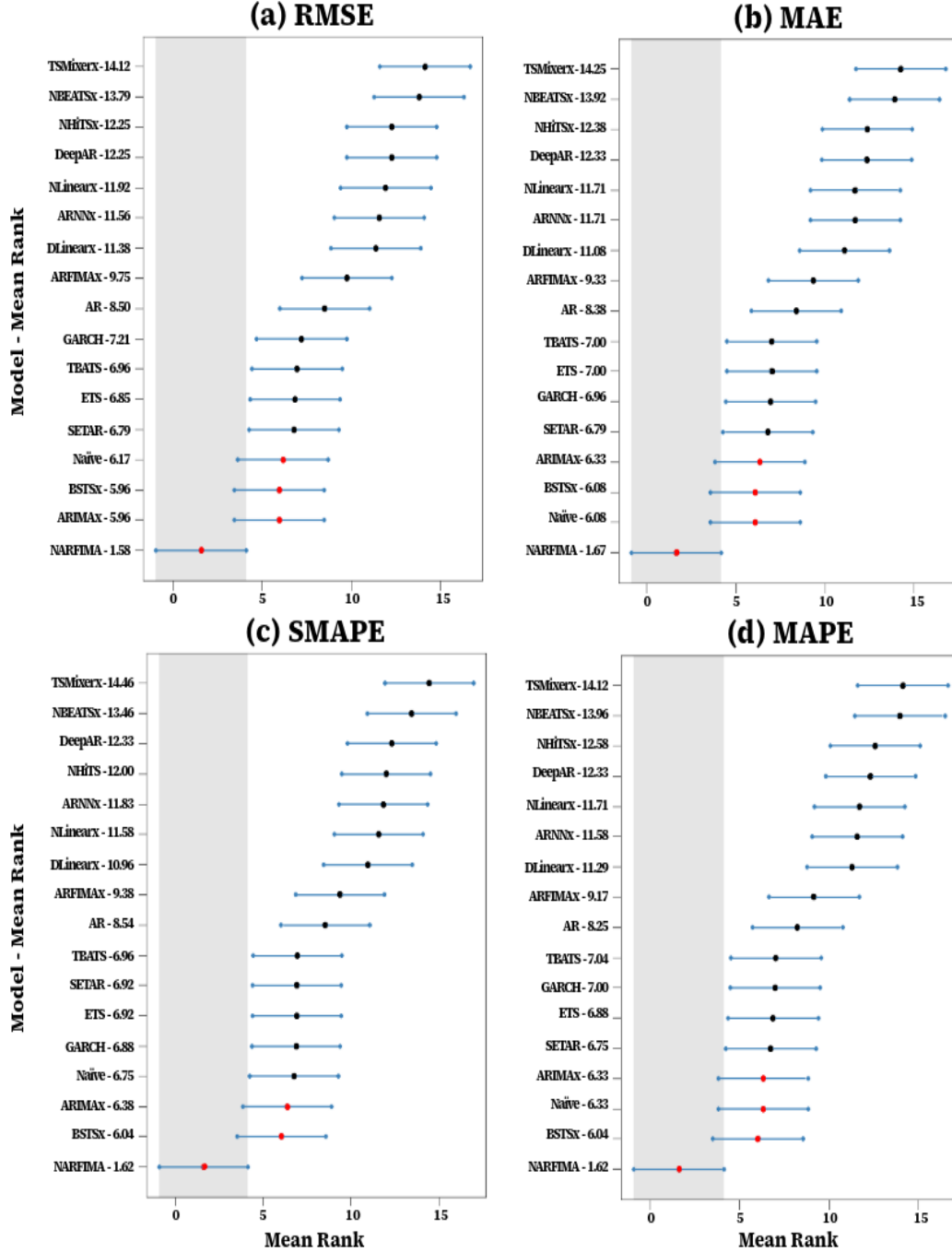


Figure 1. Multiple comparisons with the best (MCB) plot for BRIC nations based on (a) RMSE, (b) MAE, (c) SMAPE, and (d) MAPE metrics. In the plots, ‘NARFIMA-1.58’ indicates that the average rank of the NARFIMA model is 1.58, based on the RMSE metric. A similar interpretation holds across different models and metrics.

Using these weight-adjusted quantiles, the conformal prediction interval at time step t with $100(1 - \alpha)\%$ confidence is given by $[\text{NARFIMA}(\tilde{x}_t) \pm \text{CQ}_t \hat{\Psi}(\tilde{x}_t)]$. Fig. 3 represents the 95% conformal prediction intervals of the NARFIMA framework for the long-term 48-month-ahead forecast horizon. The figure depicts the point forecasts of the exchange rate series generated by the NARFIMA model and the two best-performing benchmarks, namely ARIMAX and BSTSX, as identified by RMSE-based MCB test results (Fig. 1). From the plots, it is evident that the proposed architecture can better capture the fluctuations in the exchange rate dynamics of the BRIC economies in comparison to the ARIMAX and BSTSX models. Furthermore, the width of conformal prediction intervals for the NARFIMA model varies across different countries, although it can still consistently capture most of the variations in the exchange rate datasets. However, for Brazil and Russia, both point forecasts and prediction intervals of NARFIMA and other leading models occasionally fail to capture the inherent dynamics of the exchange rate series, primarily due to abrupt

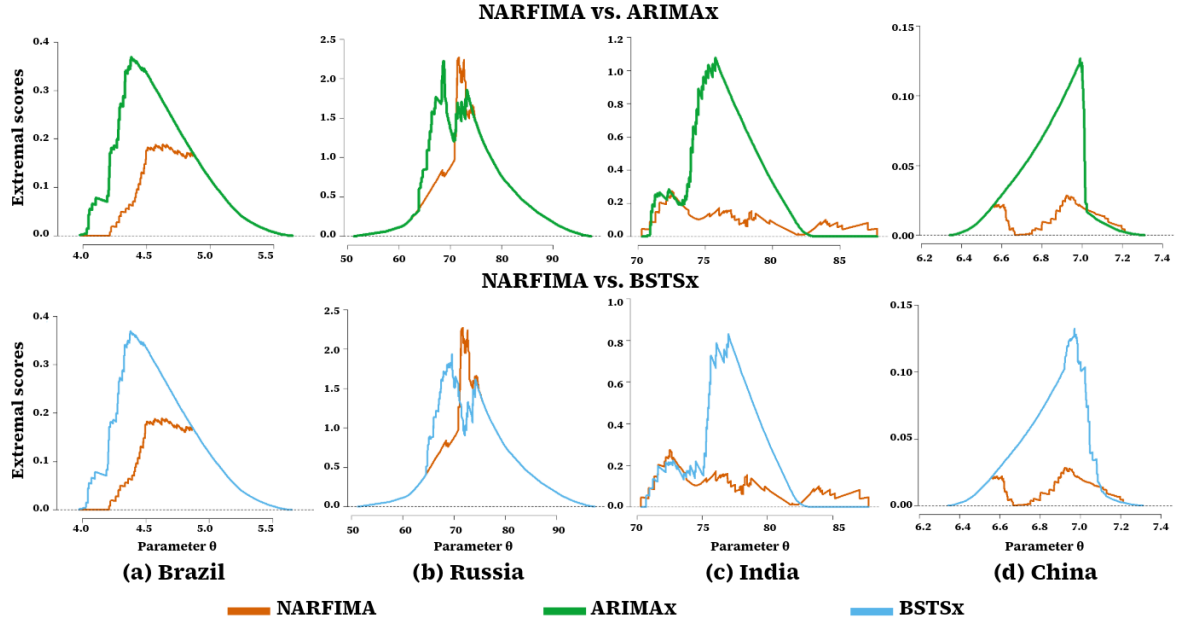


Figure 2. Murphy diagrams of NARFIMA with baselines (ARIMAX (top) and BSTSx (bottom)) for the 48-month ahead exchange rate forecasting of (a) Brazil, (b) Russia, (c) India, and (d) China. The parameter θ represents the shape parameter as defined in Eqn. (4). Lower scores indicate better performance.

macroeconomic fluctuations triggered by the COVID-19 pandemic and geopolitical conflicts, respectively. The overall analysis thus offers insights into uncertainties associated with the exchange rate forecasts for the BRIC economies.

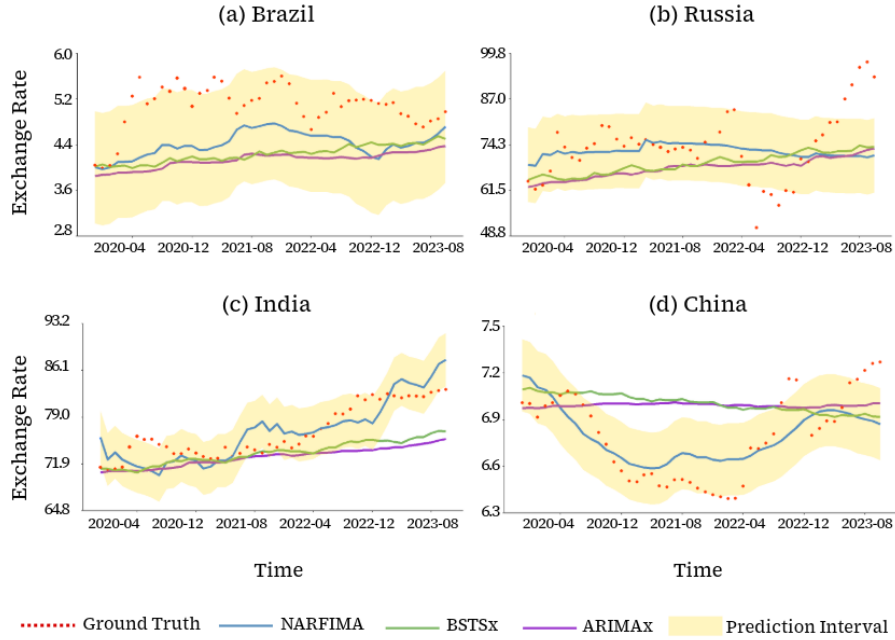


Figure 3. Visualization of the ground truth exchange rate observations (red dots) along with point forecasts from NARFIMA (blue line), BSTSx (green line), and ARIMAX (violet line), along with the conformal prediction interval for NARFIMA (yellow shaded region). Forecasts are for a 48-month-ahead horizon for (a) Brazil, (b) Russia, (c) India, and (d) China.

5.2. Sensitivity Analysis of Residual Selection

This section examines the sensitivity of the NARFIMA model to residual selection. The choice of feedback residuals plays a crucial role in designing the overall forecasting framework, as it directly influences how well long-term dependencies are captured. In

the proposed NARFIMA architecture, long-memory modeling is incorporated into the neural network by utilizing the residuals of the ARFIMAx model, along with exchange rate series and macroeconomic covariates. To empirically validate the importance of ARFIMAx residuals, we conduct an iterative forecast evaluation in two ways. First, we replace them with residuals from ARIMAx, BSTSx, and Naïve, the three best-performing benchmark models identified in the MCB plot, while keeping the neural network structure unchanged. This results in three NARFIMA variants, namely Neural ARIMAx (NARIMA), Neural BSTSx (NBSTS), and Neural Naïve (NNAïve), respectively. Second, we examine a residual-free variant of NARFIMA, identical to ARNNx, where the neural network is trained solely on exchange rate series and macroeconomic covariates, removing residual feedback altogether. This empirical evaluation aims to assess how residual selection affects overall forecast accuracy. The MCB plot in Fig. 4 demonstrates the statistical significance of the performance improvement among NARFIMA and its variants in terms of the RMSE metric. As evident from the plot, NARFIMA achieves the lowest average rank across all BRIC economies and forecast horizons, further validating the superiority of our proposal over its variants. These findings confirm that the forecasting performance of NARFIMA relies on both the neural network and ARFIMAx residuals, which optimizes its modeling capabilities.

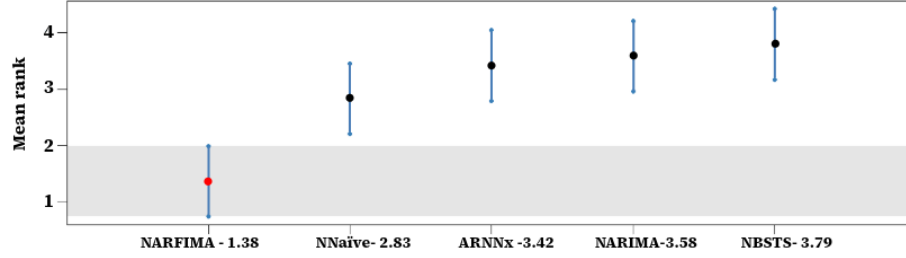


Figure 4. MCB plot comparing the performance of NARFIMA and its variants based on RMSE metrics. In the plot, ‘NARFIMA - 1.38’ indicates that the average rank of NARFIMA is 1.38, similar for other models.

6. Policy Implications

Accurate forecasting of spot exchange rates is of immense importance for the central banks, especially within emerging market economies such as BRIC. Exchange rate movements substantially influence macroeconomic stability through multiple channels, including trade competitiveness, inflation dynamics, capital flows, and the effectiveness of monetary policy. Given the varied exchange rate regimes among the BRIC economies, from managed floats to more flexible arrangements, precise forecasting is essential for mitigating external shocks, guiding foreign exchange interventions, and enhancing overall economic resilience. The recent acceleration of de-dollarization efforts in the BRIC economies, which aim to reduce dependence on the USD in trade, reserves, and settlement, has further intensified the complexity of exchange rate dynamics by introducing new drivers alongside traditional macroeconomic determinants. This shift toward greater currency diversification in trade and reserves increases the need for models that can simultaneously capture long-term dollar spillover effects and emerging local currency influences. In particular, the BRIC economies deepen their integration within global trade and financial networks while simultaneously seeking greater monetary sovereignty. From the central banks’ standpoint, improved accuracy in exchange rate forecasts enables enhanced management of foreign exchange reserves, smoother market interventions, and more effective transmission of monetary policy. Reliable forecasts also equip policymakers to proactively address capital flow volatility, assess external vulnerabilities accurately, and adjust interest rates strategically to maintain economic stability.

Given these insights, BRIC central banks must adapt their policy frameworks and exchange rate forecasting models to reflect de-dollarization dynamics. As BRIC nations increase trade settlement in local currencies, forecasting models must account for shifting demand patterns for BRIC currencies. The persistence of partial dollarization requires the models to incorporate both dollar-based and local currency-based trade flows, ensuring an accurate representation of exchange rate drivers. While the BRIC economies are reducing their reliance on dollar reserves, empirical data suggest that the USD still dominates global reserves. This means that exchange rate forecasting models must still incorporate US monetary policy spillovers, even as BRIC central banks increase holdings in gold, yuan, and other non-dollar assets. This aligns with empirical evidence underscoring the pivotal role of uncertainty measures in exchange rate forecasting, particularly within emerging markets pursuing de-dollarization. A recent study by [Abid, 2020](#) confirms that heightened EPU induces short-run and long-run depreciation pressures on emerging market currencies, suggesting that integrating EPU into forecasting models becomes even more critical during transitions away from dollar dependence. Further, [Christou et al., 2018](#) highlights the predictive power of EPU on exchange rate volatility under extreme market conditions, underscoring its relevance in forecasting strategies during periods of monetary transition. The significant impact of US MPU on global currency volatility, as highlighted by [Mueller et al., 2017](#), reinforces the necessity for the BRIC economies to incorporate MPU into predictive frameworks, particularly as they attempt to reduce vulnerability to Fed policy shifts through currency diversification. The causal relationships identified in Table 2 affirm the importance of these uncertainty indicators as causal drivers for BRIC exchange rates.

In this context, the proposed NARFIMA model provides a robust framework capable of dynamically capturing the complex influences of uncertainty measures, country-specific short-term IRD, and oil shocks on exchange rates. The model’s capacity to incorporate these key causal drivers is particularly valuable as the BRIC economies navigate the transition toward de-dollarization,

where currency valuations become increasingly sensitive to both global uncertainties and structural shifts in international monetary arrangements. Additionally, our proposed NARFIMA model can address nonlinearities, long memories, and structural breaks inherent in de-dollarization processes, equipping central banks with a powerful tool to anticipate exchange rate movements in an environment characterized by evolving currency preferences and settlement mechanisms. Although initiatives such as the BRICS Contingent Reserve Arrangement and expanded currency swap agreements can help mitigate exchange rate volatility, the absence of a unified BRIC currency system means that fluctuations across national currencies continue to affect inter-BRIC trade. Finally, the study underscores the need for policy coordination among BRIC nations in the face of shared vulnerabilities to oil shocks and global uncertainty. A forecasting tool like NARFIMA that incorporates both domestic fundamentals and global risk drivers can serve as a basis for joint monitoring mechanisms, facilitating greater financial cooperation and resilience building within the bloc.

7. Conclusion and Discussion

This paper introduces a Neural ARFIMA model to accurately forecast the spot exchange rate dynamics in BRIC countries by taking into account various uncertainty measures, oil shocks, and country-specific short-term IRD. Our proposed approach effectively combines the memory properties of fractionally integrated processes with the flexible nonlinear mapping capabilities of neural networks, creating a powerful methodological solution for capturing both long-range dependencies and complex nonlinear patterns in exchange rate data. Our empirical results across BRIC economy exchange rates demonstrate that the proposed NARFIMA model consistently outperforms traditional statistical and state-of-the-art deep learning approaches across various forecasting horizons. The conditions for asymptotic stationarity and geometric ergodicity of the NARFIMA process are established by analyzing the asymptotic behavior of the associated Markov chain. Under appropriate parameter constraints on skip connections and activation function characteristics, we prove that the NARFIMA process converges to a unique stationary distribution at a geometric rate, regardless of initial conditions. This mathematical framework provides critical assurance regarding the model's long-term behavior, validating its application to exchange rate forecasting problems where stability and convergence are essential properties. The nonlinear GC test depicted the complex interplay between key macroeconomic drivers and exchange rate series. By focusing on spot exchange rates rather than real effective exchange rates, our framework targets the price directly faced by market participants in trade, investment, and policy operations, ensuring that the forecasts are immediately relevant for day-to-day decision-making in foreign exchange management. The NARFIMA model, therefore, offers significant practical value for central banks and financial institutions engaged in monetary policy formulation and foreign exchange operations, particularly during periods of heightened uncertainty. While our current implementation focuses on the BRIC economies and incorporates several key uncertainty measures, future research could extend the NARFIMA framework to include additional factors such as climate risk indicators and social media-based uncertainty measures derived from platforms like Twitter (X). Furthermore, alternative neural network architectures could be explored within the NARFIMA structure to potentially enhance modeling flexibility and forecasting accuracy. The model could also be adapted to forecast other financial indicators characterized by long memory and nonlinear dynamics, including interest rates, commodity prices, and equity market volatility. Given its robust theoretical foundations and empirical performance, NARFIMA presents a promising direction for advancing our understanding of complex economic relationships in an increasingly interconnected global economy. Although the NARFIMA model is primarily developed for long memory and nonlinear time series problems arising in BRIC exchange rate series, it can also be useful for similar complex temporal data problems arising in epidemiology, demand forecasting, and climatology.

Competing interests

There are no competing interests to be declared.

Data and Code Availability Statement

The spot exchange rate, along with all macroeconomic indicators used in this analysis, is obtained from the Federal Reserve Economic Data (FRED) repository: <https://fred.stlouisfed.org>. Data for all uncertainty indicators are sourced from the Economic Policy Uncertainty website: <https://www.policyuncertainty.com>. The code and data necessary to reproduce the results of this study are available at: <https://github.com/mad-stat/NARFIMA>.

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A. Proofs of the Theoretical Results

A.1. Proof of Lemma 1

Proof Assumption (2) holds for the proposed NARFIMA model by design because Assumption (2) is satisfied by the logistic activation function. Given this, it follows that for any scalars β_0, β_i, μ_i , and $\phi_i \neq 0 \forall i \in 1, \dots, k$, the condition $\beta_0 + \sum_{i=1}^k \beta_i G(\phi_i x + \mu_i) = 0$, implies that $\beta_0 = \beta_1 = \dots = \beta_k = 0 \forall x \in \mathcal{R} \forall k \in \mathcal{Z}^+$. Consider the generalized controllability matrix, which is defined as:

$$C'_{x_0} = \begin{bmatrix} c & 1 \\ 0 & 1 \end{bmatrix},$$

where $c = \psi_1 + \psi_2 + \sum_{i=1}^k \beta_i (\phi_{i,1} + \phi_{i,2}) G((\phi_{i,1} + \phi_{i,2})e_1 + \phi_{i,1}\hat{y}_1 + \mu_i)$. It can be established that for any choice of $\hat{y}_1$, there exists some $e_1 \in \theta$ such that $c \neq 0$. By setting $\theta \equiv \mathcal{R}$ and selecting $\hat{y}_1$ appropriately, Assumption (3) in Lemma 1 ensures that c is nonzero for at least one $e_1 \in \theta$. Consequently, C'_{x_0} is non-singular, which confirms that the control system defined by Eqn. (3) is forward accessible. $\square$

A.2. Proof of Lemma 2

Proof Fix $y \in \mathcal{R}$ and let $O \subset \mathcal{R}$ be a nonempty open interval. Conditionally on $(y_{t-1}, e_{t-1}) = (y, e)$,

$$y_t = f(y, e) + \varepsilon_t; \quad f(y, e) := \psi_1 y + \psi_2 e + g(y, e).$$

By Assumption (4), ε_t has a strictly positive continuous density on $\mathcal{R}$. Therefore,

$$\mathbb{P}(y_t \in O \mid y_{t-1} = y, e_{t-1} = e) = \int_O f_\varepsilon(y' - f(y, e)) dy' > 0, \quad \forall y' \in \mathcal{R}.$$

To show that the conditional one-step transition kernel in y has a strictly positive density everywhere, we remove the conditioning on e_{t-1} :

$$\mathbb{P}(y_t \in O \mid y_{t-1} = y) = \mathbb{E} \left[\int_O f_\varepsilon(y' - f(y, e_{t-1})) dy' \mid y_{t-1} = y \right].$$

By Assumption (4) the inner integral is strictly positive for every realization of e_{t-1} since $f_\varepsilon > 0$ everywhere. Hence, the conditional expectation is also strictly positive. Therefore, from any y , every nonempty open set O is reached in one step with positive probability. This establishes irreducibility. If we take O to be any neighborhood of y , then $\mathbb{P}(y_t \in O \mid y_{t-1} = y) > 0$, which rules out cyclic behavior and yields aperiodicity. $\square$

A.3. Proof of Theorem 1

Proof In order to verify the geometric drift condition, we choose the general Lyapunov function $V(y) = 1 + y^2$. Then,

$$\mathbb{E}[V(y_t) \mid y_{t-1} = y, e_{t-1} = e] = 1 + \mathbb{E}[(\psi_1 y + \psi_2 e + g(y, e) + \varepsilon_t)^2] = 1 + (\psi_1 y + \psi_2 e + g(y, e))^2 + \mathbb{E}(\varepsilon_t^2).$$

$\forall \eta > 0$ the bound $(a + b + c)^2 \leq (1 + \eta)a^2 + \left(2 + \frac{2}{\eta}\right)(b^2 + c^2)$ holds $\forall a, b, c \in \mathcal{R}$ and by Assumption (2) $\exists A > 0; |g(y, e)| \leq A$. Thus,

$$(\psi_1 y + \psi_2 e + g)^2 \leq (1 + \eta)\psi_1^2 y^2 + \left(2 + \frac{2}{\eta}\right)(\psi_2^2 e^2 + A^2).$$

Now, we can fix η such that $(1 + \eta)\psi_1^2 = 1 - \delta$ for some $\delta \in (0, 1)$ by Assumption (5). This yields,

$$\mathbb{E}[V(y_t) \mid y_{t-1} = y, e_{t-1} = e] \leq 1 + (1 - \delta)y^2 + \left[\left(2 + \frac{2}{\eta}\right)A^2 + \mathbb{E}(\varepsilon_t^2)\right] + \left(2 + \frac{2}{\eta}\right)\psi_2^2 e^2.$$

Notice that $1 + (1 - \delta)y^2 = (1 - \delta)V(y) + \delta$. Hence,

$$\mathbb{E}[V(y_t) \mid y_{t-1} = y, e_{t-1} = e] \leq (1 - \delta)V(y) + \underbrace{\left[\delta + \left(2 + \frac{2}{\eta}\right)A^2 + \mathbb{E}(\varepsilon_t^2)\right]}_{=: b_0} + \underbrace{\left(2 + \frac{2}{\eta}\right)\psi_2^2 e^2}_{=: b_1}.$$

Finally, by Assumptions (1) and (4) (finite variance of ε_t), we have that

$$\mathbb{E}[V(y_t) \mid y_{t-1} = y] \leq (1 - \delta)V(y) + B, \quad B := b_0 + b_1 e^2 < \infty.$$

This is a Foster-Lyapunov drift (Xu et al., 2018). This geometric drift condition, combined with irreducibility, results in geometric ergodicity. Therefore, $\{y_t\}$ admits a unique invariant distribution and converges to it at a geometric rate. If the initial draw (or sample) y_0 is drawn from the invariant law, the process is asymptotically stationary. $\square$

B. Macroeconomic Datasets and Global Characteristics

This section provides a brief overview of the uncertainty measures used in our analysis and describes the global features of all the macroeconomic variables.

B.1. Overview Uncertainty Measures

In this study, we consider four newspaper-based uncertainty measures, namely GEPU, US EMV, US MPU, and GPR index. The GEPU index, developed by [Davis, 2016](#), reflects global economic uncertainty that impacts investment, trade, and financial markets. It is computed as a GDP-weighted average of national EPU indices from 21 countries, which collectively account for two-thirds of global output. The national EPU index measures uncertainty related to economic policy decisions, their timing, impact, and the economic consequences of non-economic events such as military actions ([Baker et al., 2016](#)). It is constructed by analyzing the frequency of terms related to economy, uncertainty, and policy in country-specific newspaper articles. The GEPU index, derived from these national EPU indices using PPP-adjusted GDP, effectively captures both global economic crises and geopolitical disruptions. The US EMV index, introduced by [Baker et al., 2019](#), quantifies the impact of wars, policy risks, commodity markets, and macroeconomic outlook on US equity returns and stock market volatility. It tracks the relative frequency of terms related to the economy, stock markets, and volatility across eleven major US newspapers. By leveraging electronic text search techniques, it helps quantify fluctuations in the CBOE Volatility Index and the realized volatility of S&P 500 returns. This index reflects how policy uncertainty, market sentiment, and global economic shocks influence market fluctuations and investor behavior in the US financial markets. The US MPU index, proposed by [Husted et al., 2020](#), specifically measures uncertainty surrounding Federal Reserve monetary policy actions and their implications. It bridges the gap between conventional and unconventional policy regimes by tracking the frequency of articles containing monetary policy, uncertainty, and Federal Reserve-related terms in the Washington Post, Wall Street Journal, and New York Times. Unlike the GEPU and US EMV indices, which capture broader uncertainty sources such as fiscal policy, healthcare, and national security, the US MPU index focuses solely on the US monetary policy landscape, reflecting the time-varying influence of Federal Reserve communications. Alongside economic uncertainties, we also analyze the country-specific GPR index, developed by [Caldara and Iacoviello, 2022](#), which quantifies risks arising from adverse geopolitical events such as terrorism, political instability, violence, territorial conflicts, and wars. This country-specific index is constructed using electronic text searches of relevant geopolitical terms across ten prominent newspapers from the US, UK, and Canada. It captures global perceptions of geopolitical risks associated with a given country or its major cities.

B.2. Statistical Properties of the Macroeconomic Time Series

We compute the five-point summary, mean, standard deviation (SD), coefficient of variation (CoV), and entropy for all the macroeconomic variables and uncertainty measures used in our analysis. The values of these summary statistics, reported in Table 10, offer valuable economic insights. For instance, the CoV reveals substantial relative variability across most economic indicators, with notable differences between countries and variables. Generally, exchange rates exhibit moderate CoV values, whereas interest rate differentials and inflation differentials show much higher relative variability. Alongside the descriptive statistics, we analyze the global behavior of the exchange rate series and exogenous variables, and summarize the results in Table 11. The analysis includes key features such as skewness, kurtosis, nonlinearity, seasonality, stationarity, long-range dependence, and detected outliers, offering a comprehensive overview of the structural patterns across all time series.

Table 10. Summary statistics of the datasets utilized in this analysis.

Country	Series	Min Value	Q1	Median	Mean	SD	Q3	Max Value	CoV	Entropy
Brazil	Exchange Rate	1.043	1.800	2.215	78.611	2.382	3.004	4.120	33.002	5.608
	Short-term Interest Rate	5.480	10.578	13.750	15.534	814.479	18.643	49.750	52.432	4.728
	Short-term IRD	3.650	8.798	11.720	13.291	721.781	15.523	44.680	54.306	5.425
	CPI Inflation	1.645	4.518	5.989	6.203	270.569	7.259	17.236	43.619	5.613
	CPI Inflation Differential	-0.290	1.925	3.487	4.050	297.201	5.403	15.178	73.383	5.613
	GPR	0.003	0.026	0.039	0.048	3.365	0.059	0.225	70.107	5.613
Russia	Exchange Rate	5.629	27.624	29.899	34.673	1632.301	35.372	75.172	47.077	5.608
	Short-term Interest Rate	5.250	8.250	11.000	17.316	1632.830	21.000	150.000	94.296	3.432
	Short-term IRD	4.670	6.368	9.600	15.073	1514.957	18.475	144.510	100.508	5.199
	CPI Inflation	2.197	6.853	10.232	15.022	1974.837	14.858	126.512	131.463	5.613
	CPI Inflation Differential	-0.572	4.959	7.697	12.868	1977.207	13.297	124.367	153.653	5.613
	GPR	0.205	0.394	0.550	0.652	34.814	0.814	2.329	53.396	5.613
India	Exchange Rate	35.747	43.924	46.787	50.983	992.537	60.935	73.561	19.468	5.613
	Short-term Interest Rate	5.400	6.000	6.500	7.161	142.468	8.188	12.000	19.895	2.301
	Short-term IRD	0.470	3.453	4.875	4.919	233.033	5.920	10.170	47.374	5.010
	CPI Inflation	0.000	4.167	5.932	6.675	334.077	8.824	19.672	50.049	5.506
	CPI Inflation Differential	-2.622	1.444	3.932	4.521	381.554	6.785	18.124	84.396	5.613
	GPR	0.044	0.133	0.181	0.213	14.577	0.249	1.126	68.436	5.613
China	Exchange Rate	6.051	6.570	7.068	7.345	85.356	8.277	8.326	11.621	5.157
	Short-term Interest Rate	2.700	2.900	3.240	3.396	123.672	3.250	9.000	36.417	1.164
	Short-term IRD	-5.260	-2.118	-0.270	187.761	-0.904	-0.090	3.810	-207.700	4.639
	CPI Inflation	-2.200	0.784	1.781	1.953	213.111	2.816	8.805	109.120	5.601
	CPI Inflation Differential	-4.477	-1.823	-0.019	-0.200	192.720	1.038	4.778	-963.602	5.613
	GPR	0.098	0.296	0.389	0.456	23.725	0.540	1.521	52.028	5.613
Global	GEPU	51.141	76.741	104.319	117.880	5429.789	145.306	337.760	46.062	5.613
	US EMV	9.570	15.529	18.854	21.144	827.153	24.324	69.835	39.120	5.613
	US MPU	19.749	74.521	103.705	114.952	6343.051	134.119	407.365	55.180	5.613
	Oil Price Growth Rate	-59.021	-12.328	7.734	10.212	3650.524	31.315	144.605	357.474	5.613
	US Short-term Interest Rate	0.070	0.160	1.650	2.243	214.821	4.725	6.540	95.774	4.638
	US CPI Inflation	-2.097	1.551	2.098	2.154	118.125	2.908	5.600	54.840	5.608

Table 11. Global characteristics of the economic time series under study for BRIC countries.

Countries	Series	Skewness	Kurtosis	Nonlinearity	Seasonality	Stationarity	Hurst Exponent	Outlier(s) Detected
Brazil	Exchange Rate	0.417	-0.689	Non-linear	Non-seasonal	Non-stationary	0.784	1
	Short-term Interest Rate	1.820	3.970	Non-linear	Non-seasonal	Non-stationary	0.816	5
	Short-term IRD	1.864	4.343	Nonlinear	Non-seasonal	Non-stationary	0.796	2
	CPI Inflation	1.540	3.778	Nonlinear	Non-seasonal	Stationary	0.735	4
	CPI Inflation Differential	1.233	1.842	Linear	Non-seasonal	Stationary	0.690	1
	GPR	2.313	7.506	Nonlinear	Non-seasonal	Stationary	0.641	5
Russia	Exchange Rate	0.730	-0.0378	Nonlinear	Non-seasonal	Non-stationary	0.824	0
	Short-term Interest Rate	3.230	16.560	Nonlinear	Non-seasonal	Non-stationary	0.804	2
	Short-term IRD	3.552	20.162	Nonlinear	Non-seasonal	Non-stationary	0.793	2
	CPI Inflation	4.042	16.931	Nonlinear	Non-seasonal	Non-stationary	0.745	8
	CPI Inflation Differential	4.048	16.957	Nonlinear	Non-seasonal	Non-stationary	0.743	8
	GPR	2.056	5.412	Nonlinear	Non-seasonal	Stationary	0.622	4
India	Exchange Rate	0.683	-0.853	Linear	Non-seasonal	Non-stationary	0.848	1
	Short-term Interest Rate	1.098	0.432	Nonlinear	Non-seasonal	Non-stationary	0.817	1
	Short-term IRD	0.100	-0.604	Nonlinear	Non-seasonal	Non-stationary	0.826	1
	CPI Inflation	0.921	0.938	Nonlinear	Non-seasonal	Stationary	0.768	1
	CPI Inflation Differential	0.792	0.311	Nonlinear	Non-seasonal	Stationary	0.786	1
	GPR	3.246	13.617	Nonlinear	Non-seasonal	Non-stationary	0.704	6
China	Exchange Rate	-0.0384	-1.727	Nonlinear	Non-seasonal	Non-stationary	0.868	1
	Short-term Interest Rate	3.521	12.218	Nonlinear	Non-seasonal	Non-stationary	0.696	4
	Short-term IRD	-0.469	0.193	Nonlinear	Non-seasonal	Stationary	0.730	1
	CPI Inflation	0.617	0.569	Linear	Non-seasonal	Non-stationary	0.763	1
	CPI Inflation Differential	0.0189	-0.536	Linear	Non-seasonal	Non-stationary	0.807	1
	GPR	1.568	2.757	Linear	Non-seasonal	Non-stationary	0.730	1
Global	GEPU	1.374	1.884	Nonlinear	Seasonal	Non-stationary	0.806	1
	US EMV	2.196	7.213	Nonlinear	Seasonal	Non-stationary	0.698	4
	US MPU	1.874	4.761	Nonlinear	Non-seasonal	Non-stationary	0.706	3
	Oil Price Growth Rate	0.600	0.556	Nonlinear	Non-seasonal	Stationary	0.723	1
	US Short-term Interest Rate	0.573	-1.235	Nonlinear	Non-seasonal	Non-stationary	0.830	1
	US CPI Inflation	-0.307	1.056	Nonlinear	Non-seasonal	Non-stationary	0.762	1

C. Empirical Setup and Results

This section provides an overview of the state-of-the-art forecasting models used for benchmarking, in addition to, the empirical evidence of the nonlinearity of ARFIMAx’s residuals.

C.1. Overview of the Baseline Models

We evaluate the performance of the proposed NARFIMA model against several state-of-the-art models to demonstrate its forecasting effectiveness. The evaluation incorporates the following methods:

- *Naïve* forecasting approach predicts future values by using the last observed value of a time series without any adjustments. Despite its simplicity, this stochastic model is often effective for economic time series and serves as a benchmark for evaluating the performance of more advanced forecasting techniques (Hyndman and Athanasopoulos, 2018).
- *Autoregressive* (AR) model assumes that future values of a time series are primarily influenced by its past observations. It closely resembles multiple linear regression, except that the target series is predicted using its own lagged values. This method is most suitable for modeling stationary time series datasets (Hyndman and Athanasopoulos, 2018).
- *Autoregressive Integrated Moving Average with exogenous variables* (ARIMAx) model is a widely used time series forecasting technique that captures linear dependencies by combining three key components: autoregressive (AR), differencing (I), and moving average (MA) (Box and Pierce, 1970). In this framework, differencing of order d_0 ensures the stationarity of the input series, after which the AR component models the p_0 lagged values of the series, while the MA component incorporates q_0 lagged residuals. The coefficients of ARIMAx are estimated by optimizing the Akaike information criterion (AIC).
- *Autoregressive Fractionally Integrated Moving Average with exogenous variables* (ARFIMAx) extends ARIMAx by allowing the differencing parameter to take fractional values in the range $(0, 0.5)$. This generalization enables the ARFIMAx($\tilde{p}, d, \tilde{q}$) model to effectively capture long-range dependencies in time series data, where correlations decay gradually (Granger and Joyeux, 1980). As shown in Table 11, the exchange rate series exhibits long memory dynamics, justifying the use of the ARFIMAx model for accurate forecasting.
- *Exponential Smoothing* (ETS) model decomposes a time series into three components: noise, trend, and seasonal patterns. Each component can be modeled using either an additive or multiplicative approach. ETS estimates its parameters by minimizing the sum of squared errors and is known for its flexibility in producing accurate forecasts across a wide range of time series data (Hyndman and Athanasopoulos, 2018).
- *Self-exciting Threshold Autoregressive* (SETAR) model extends the traditional autoregressive approach by incorporating threshold effects. It partitions the time series into distinct regimes based on a threshold value, allowing different autoregressive processes in each regime. This structure enables SETAR to capture nonlinearities and model complex time series with regime-switching behavior (Firat, 2017).
- *Trigonometric Box-Cox ARIMA Trend Seasonal* (TBATS) model is designed for time series with multiple seasonal patterns and nonlinear trends. TBATS enhances traditional models by incorporating Fourier terms, applying a Box-Cox transformation to stabilize variance, and using an ARIMA model to account for residual autocorrelation. This approach enables TBATS to handle complex seasonal structures that are often challenging for conventional models (Hyndman and Athanasopoulos, 2018).
- *Generalized Autoregressive Conditional Heteroscedasticity* (GARCH) model captures time-varying volatility in temporal data. GARCH(p_g, q_g) incorporates p_g lagged values of the conditional variance and q_g lagged values of the squared residuals to model the volatility clustering commonly observed in financial time series. The parameter estimation is typically performed using maximum likelihood estimation. Due to its ability to forecast volatility, GARCH is widely used in econometric and financial modeling (Bollerslev, 1986).

- *Bayesian Structural Time Series with exogenous variables* (BSTSx) model is a flexible forecasting approach that decomposes a time series into components including trend, seasonality, and the effects of external covariates. It relies on a state-space representation, where observations depend on hidden states that evolve over time. The Kalman filter is used to estimate these states, while the spike-and-slab prior helps identify relevant predictors. Markov Chain Monte Carlo methods are employed to generate samples from the model's posterior distribution, and forecasts are derived by summarizing these samples (Scott and Varian, 2014).
- *Autoregressive Neural Network with exogenous variables* (ARNNx) extends conventional feed-forward neural networks to capture autoregressive time series patterns (Faraway and Chatfield, 1998). It processes b lagged values in the input layer, passing them through c neurons in the hidden layer. To ensure model stability and restrict overfitting, $c = \lfloor \frac{b+1}{2} \rfloor$ is specified. The model is initialized with random weights and trained using the gradient descent backpropagation approach (Rumelhart et al., 1986). This training process helps prevent overfitting and ensures a stable learning structure (Hyndman and Athanasopoulos, 2018).
- *Deep learning-based Autoregressive* (DeepAR) model is an advanced neural network approach designed for time series forecasting. It leverages a recurrent neural network (RNN) architecture to capture temporal dependencies and predict future values in sequential data (Salinas et al., 2020).
- *Neural Basis Expansion Analysis for Time Series with exogenous variables* (NBeatsx) model is a specialized neural network architecture for time series forecasting. It consists of multiple stacked blocks, where each block has two key layers (Oreshkin et al., 2020). The first layer captures patterns in the data, while the second layer refines the forecast by modeling residual errors. This iterative process enhances forecasting accuracy by progressively improving predictions.
- *Neural Hierarchical Interpolation for Time Series with exogenous variables* (NHITSx) model builds on the NBeatsx framework by introducing a hierarchical approach to time series forecasting. Similar to NBeatsx, this framework utilizes a series of stacked blocks and enhances them with a hierarchical structure to better capture complex dependencies within the data (Challu et al., 2023).
- *Decomposition-based Linear model with exogenous variables* (DLinearx) employs a decomposition method to separate the input time series into trend and seasonal components using a moving average. Each component is then processed through a dedicated linear layer, and their sum generates the final forecast. By explicitly modeling trends, this method improves forecasting performance, making it more effective for capturing temporal patterns in the data (Zeng et al., 2023).
- *Normalization-based Linear model with exogenous variables* (NLinearx) applies a normalization technique to improve forecasting accuracy. It normalizes the input time series by subtracting the last observed value before passing the data through a linear layer. After processing, this value is added back to generate the final forecast. This approach helps mitigate distribution shifts between training and testing data, enhancing model robustness (Zeng et al., 2023).
- *Time Series Mixer with exogenous variables* (TSMixerx) is a neural network architecture designed for time series forecasting. It employs a sequence-mixing approach, where each layer mixes information across different time steps to capture dependencies within the data. The initial layers aggregate sequential information, while subsequent layers refine the learned patterns to improve forecasting accuracy (Chen et al., 2023).

C.2. Empirical Evidence of Nonlinearity in ARFIMA Residuals

Table 12 reports the results of the Teräsvirta and BDS tests applied to the residuals of the ARFIMAx model. These results confirm the presence of significant nonlinearity, indicating that ARFIMAx captures the linear dependencies, leaving the remaining nonlinear structure to be modeled by the feed-forward neural network component of NARFIMA.

Table 12. Terasvirta Test and the BDS Test results applied to the residuals of the ARFIMAx.

Country	Horizon	Teräsvirta Test p-value	BDS Test p-value	Conclusion
Brazil	1	1.53×10^{-3}	7.59×10^{-17}	Nonlinear Residuals
	3	1.52×10^{-3}	1.58×10^{-17}	Nonlinear Residuals
	6	1.68×10^{-3}	2.52×10^{-16}	Nonlinear Residuals
	12	1.64×10^{-3}	2.07×10^{-16}	Nonlinear Residuals
	24	1.43×10^{-3}	3.53×10^{-21}	Nonlinear Residuals
	48	1.93×10^{-3}	2.72×10^{-15}	Nonlinear Residuals
Russia	1	2.56×10^{-4}	3.58×10^{-25}	Nonlinear Residuals
	3	4.66×10^{-4}	3.39×10^{-23}	Nonlinear Residuals
	6	4.55×10^{-4}	1.83×10^{-24}	Nonlinear Residuals
	12	4.11×10^{-2}	7.14×10^{-42}	Nonlinear Residuals
	24	9.14×10^{-9}	4.90×10^{-35}	Nonlinear Residuals
	48	3.49×10^{-4}	1.27×10^{-30}	Nonlinear Residuals
India	1	5.93×10^{-6}	1.04×10^{-10}	Nonlinear Residuals
	3	8.16×10^{-6}	2.82×10^{-11}	Nonlinear Residuals
	6	1.34×10^{-5}	1.51×10^{-11}	Nonlinear Residuals
	12	1.88×10^{-5}	4.16×10^{-10}	Nonlinear Residuals
	24	5.38×10^{-5}	3.54×10^{-10}	Nonlinear Residuals
	48	5.36×10^{-4}	1.26×10^{-8}	Nonlinear Residuals
China	1	2.8×10^{-9}	3.19×10^{-50}	Nonlinear Residuals
	3	6.47×10^{-9}	1.37×10^{-44}	Nonlinear Residuals
	6	3.35×10^{-9}	2.83×10^{-44}	Nonlinear Residuals
	12	4.75×10^{-10}	4.68×10^{-63}	Nonlinear Residuals
	24	5.75×10^{-10}	1.09×10^{-50}	Nonlinear Residuals
	48	1.14×10^{-8}	0.00	Nonlinear Residuals